

Box Transporting Mechanism

Self Project
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[25M0036]



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Contents

1: MODEL SPECIFICATION	3
1.1 Pictorial Depiction	3
1.2 GEOMETRY AND MATERIAL	3
1.2.1 Dimensions of every component	3
1.2.2 Masses of every component	8
1.2.3 Moments of inertia of every component	8
1.3 Constraints on the motion of each component	9
1.4 Motion transmission	10
1.4.1 Driving component	11
1.4.2 Component which gives output of interest	11
2: KINEMATIC SIMULATION	12
2.1 Output component	12
2.1.1 Simulated and expected displacement profile of the output component	14
2.1.2 Simulated and expected velocity profile of the output component	16
2.1.3 Simulated and expected acceleration profile of the output component	18
2.1.4 Comments	20
2.2 SENSITIVITY ANALYSIS	21
2.2.1 Sensitivity Analysis due to Crank Link AB	22
2.2.1 Sensitivity analysis due to Rocker Link IJ	25
2.3 ANALYSIS OF JOINT CLEARANCE	28
3: VALIDATION OF KINEMATIC SIMULATION	32
3.1 Theoretical evaluation of kinematic	32
3.1.1 Calculating Displacement Graphically	33
3.1.2 Calculating velocity and acceleration analytically and coding	37
3.2 COMPARISON OF KINEMATIC PARAMETERS	45
3.2.1 The four values of displacement evaluated in section 3.1 superimposed in the plot of section 2.1.1	45
3.2.2 The four values of velocity evaluated in section 3.1 superimposed in the plot of section 2.1.2	46
3.2.3 The four values of acceleration evaluated in section 3.1 superimposed in the plot of section 2.1.3	47
3.2.4 COMMENTS:	48
References	48

1: MODEL SPECIFICATION

1.1: Pictorial Depiction

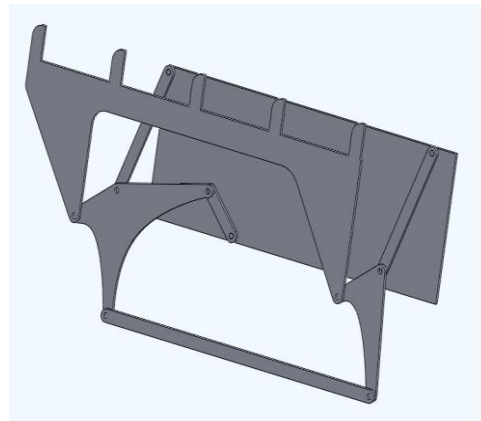


Fig.1 Actual Box Shifting Mechanism [1] and Solid works Model (Isometric view)

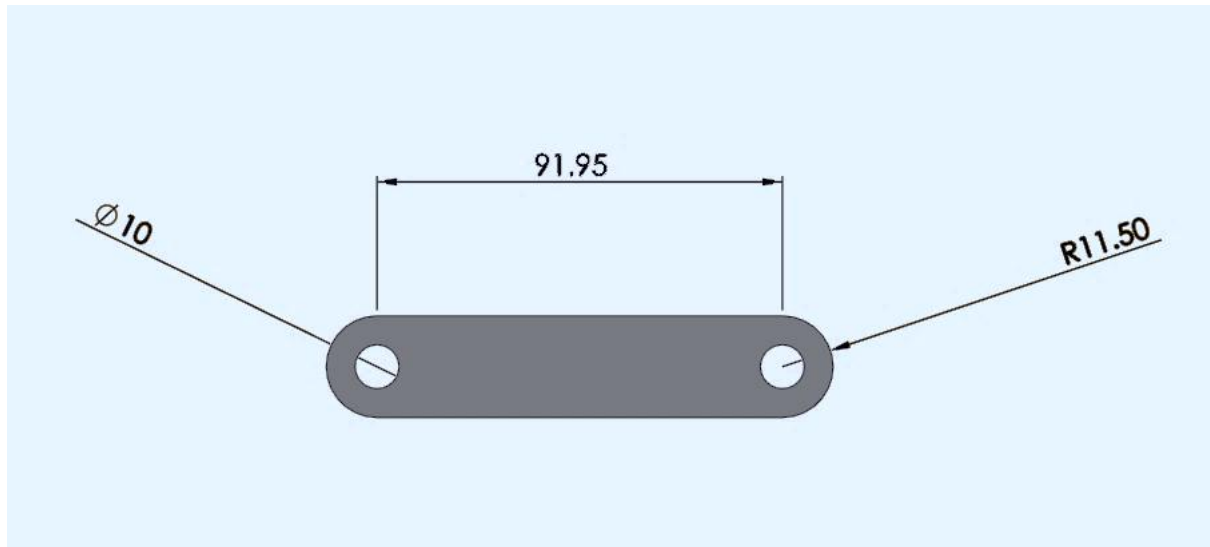
Box transport mechanism converts the rotary motion of the crank into the linear reciprocating motion of the carrier link. For small-scale businesses and internal assembly lines, it is utilised to convey boxes. The boxes move in a stop-and-move manner to its reciprocating motion.

1.2: GEOMETRY AND MATERIAL

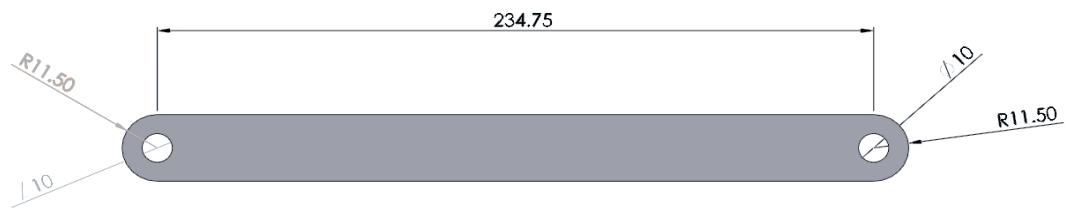
1.2.1 Dimensions of every component

All links thickness: **12.7mm** (All dimension in mm)

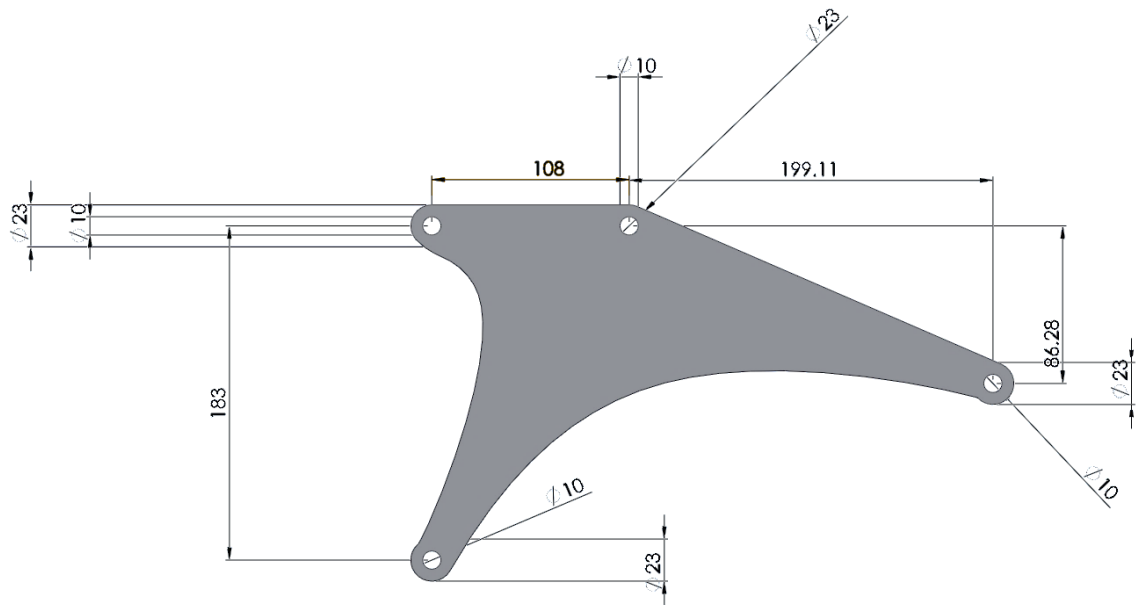
LINK AB:



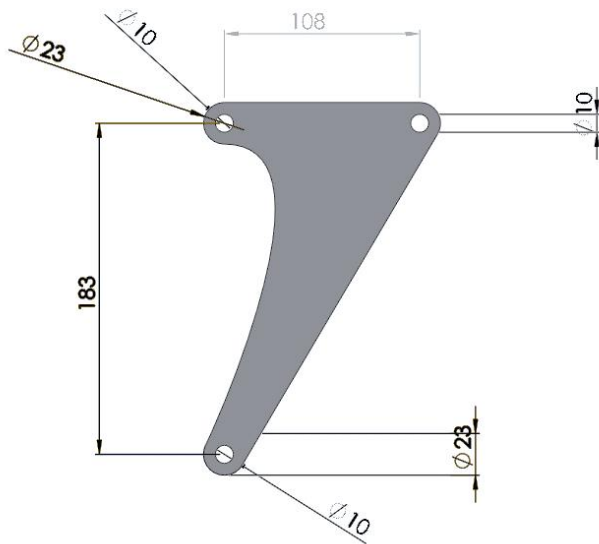
LINK CD AND IJ:



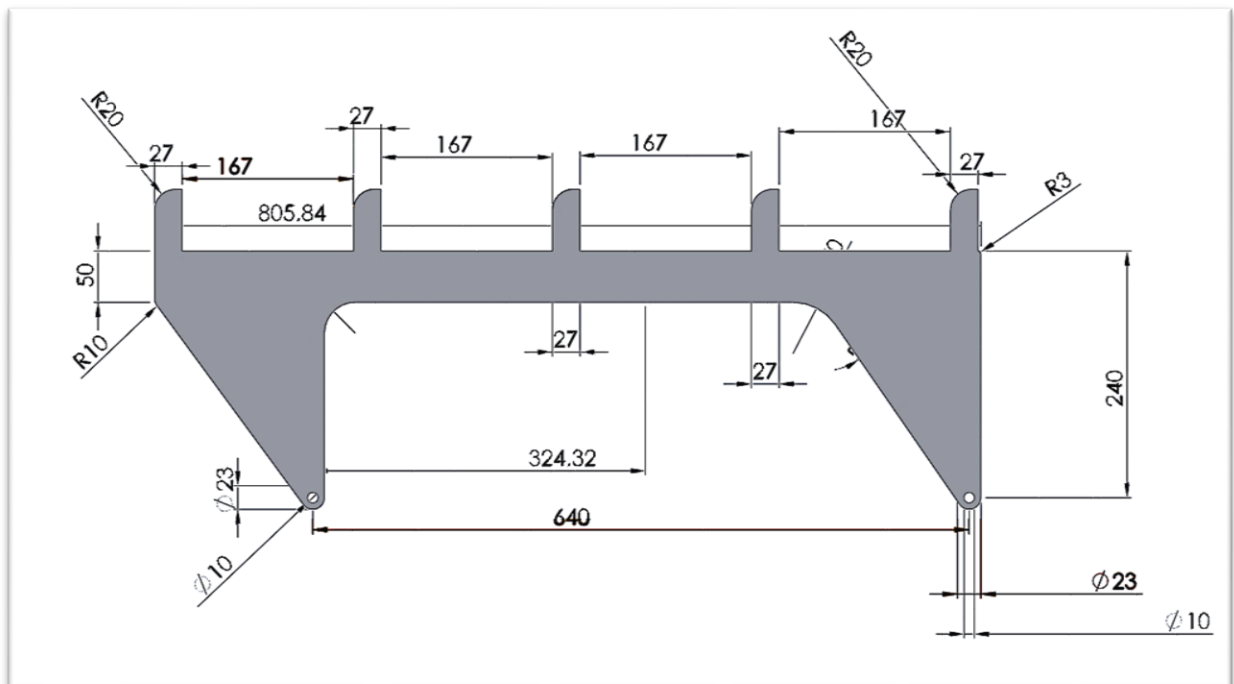
LINK BDEF:



LINK GHI:



LINK EH:



LINK FG

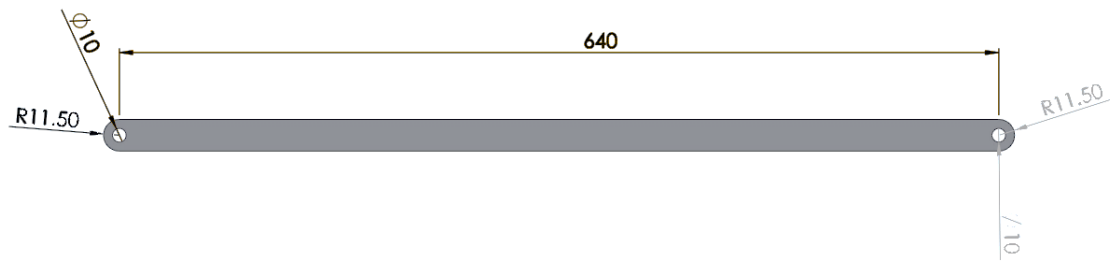


Fig.2 Dimensions of link AB, BDEF, CD, IJ, GHI, EH and FG

1.2.2 Masses of every component

Material: *DIN Steel Structure (1.0490 S275N - Solid works)*

Density: 7800 kg/m^3

Reason: For good fatigue loading

Table1: Masses of every links

Link	Name	Mass (kg)
AB	Input link/Crank	0.2350
BDEF	Quaternary link	2.40606
CD	Rocker link	0.56045
IJ	Rocker link	0.56045
GHI	Tertiary link	1.12313
EH	Output link/Carrier link	8.32971
FG	Coupler link	1.48376
ACJ	Fixed Frame/Ground	N/A

1.2.3 Moments of inertia of every component

since mechanism is planer mechanism in this all link is in parallel plane. Here I am calculating mass moment of inertia of each link about an axis perpendicular to its centre of mass. Every link has its own local coordinate system about which I_{xx} , I_{yy} and I_{zz} is calculated.

Table2: Moments of Inertia of every links

Link	I_{xx} (kg-mm ²)	I_{yy} (kg-mm ²)	I_{zz} (kg-mm ²)
AB	13.65869	224.60247	231.94144
BDEF	4309.71619	14895.53630	19140.5734
CD	32.37433	2864.78395	2882.09254
IJ	32.37433	2864.78395	2882.09254
GHI	3477.13980	948.97191	4395.92060
EH	35197.54128	647425.94855	682399.57347
FG	85.48717	52542.84125	52588.44248
ACJ	N/A	N/A	N/A

1.3: Constraints on the motion of each component

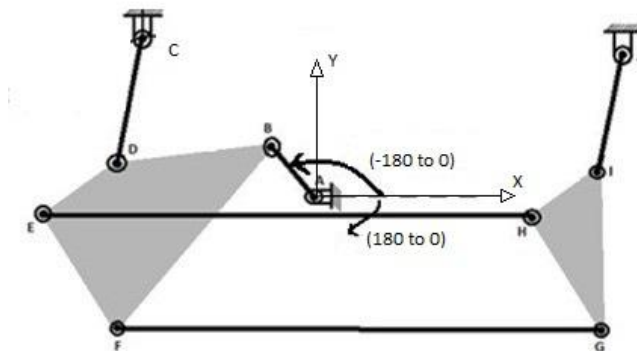


Fig. 3 Mechanism Sketch with Global Coordinate System and sign convention

Box transporting mechanism contains 8 link and 10 revolute joints connection in such manner it converts rotary motion of crank to reciprocating motion. Here all link in the parallel plane. In this mechanism there one ternary link GHI and one quaternary link BDEF and rest are binary link. and pivot point A, C, J is fixed to ground. Input of mechanism is crank AB which connected by electric motor through revolute joint. The output link EH(carrier) which has only reciprocating motion.

Link GHI is connected to link IJ, link FG, link EH by revolute joint. And link BDEF is connected to link AB, link CD, link EH, link FG by revolute joint.

Table 3: Constraints details

Link	Connected to other links	Kinematic pair/joint	Motion Of link
AB (crank)	Ground and BDEF	Revolute	Full Rotational
CD	Ground and BDEF	Revolute	Oscillatory
IJ	Ground and GHI	Revolute	Oscillatory
GHI	IJ, EH, FG	Revolute	Oscillatory
EH	BDEF and GHI	Revolute	Translational
FG	BDEF and GHI	Revolute	Translational
ACJ (Ground)	CD, IJ, AB	Revolute	Fixed

1.4: Motion transmission

$$F = 3(N-1) - 2J - H - Fr$$

F=Degree of freedom

N=Number of links =8

J= Number of lower pairs =10

H= Number of lower pairs=0

Fr= Number of redundant degrees of freedom =0

$$F = 3*(8-1) - 2*10 - 0 - 0 = 1$$

Input link: Crank (AB)

Output link: Carrier Link (EH)

1.4.1 Driving component

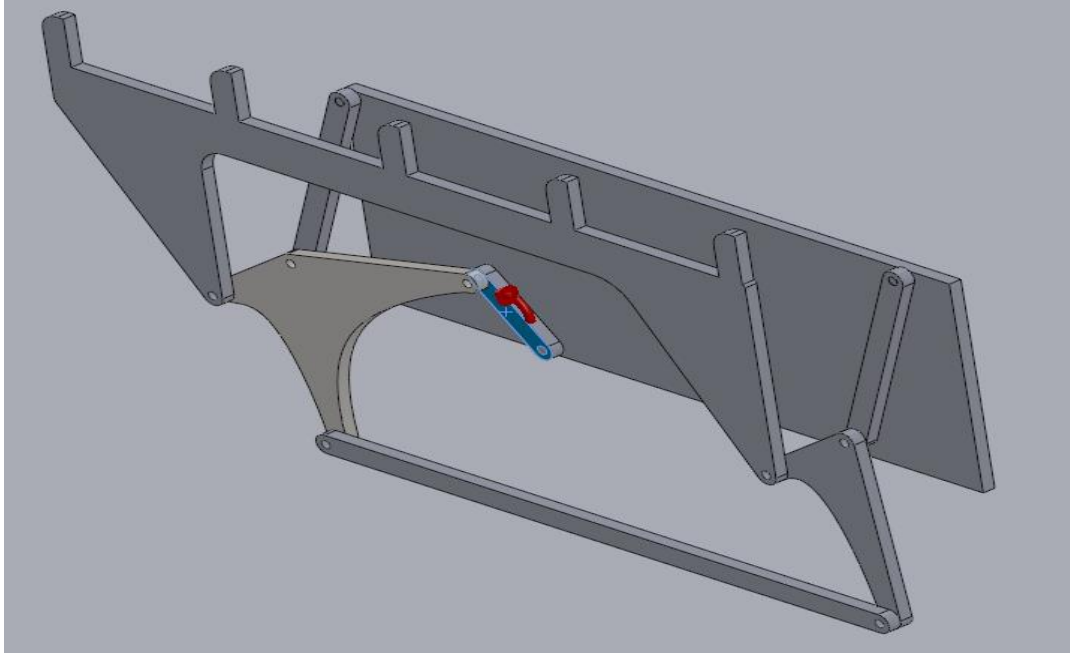


Fig.4 Solid works model showing input link.

Link AB (crank) is here driving component of mechanism, which is rotated by electric motor.

With speed of (19.098 RPM) a and it is shown in fig 4 .

1.4.2 Component which gives output of interest

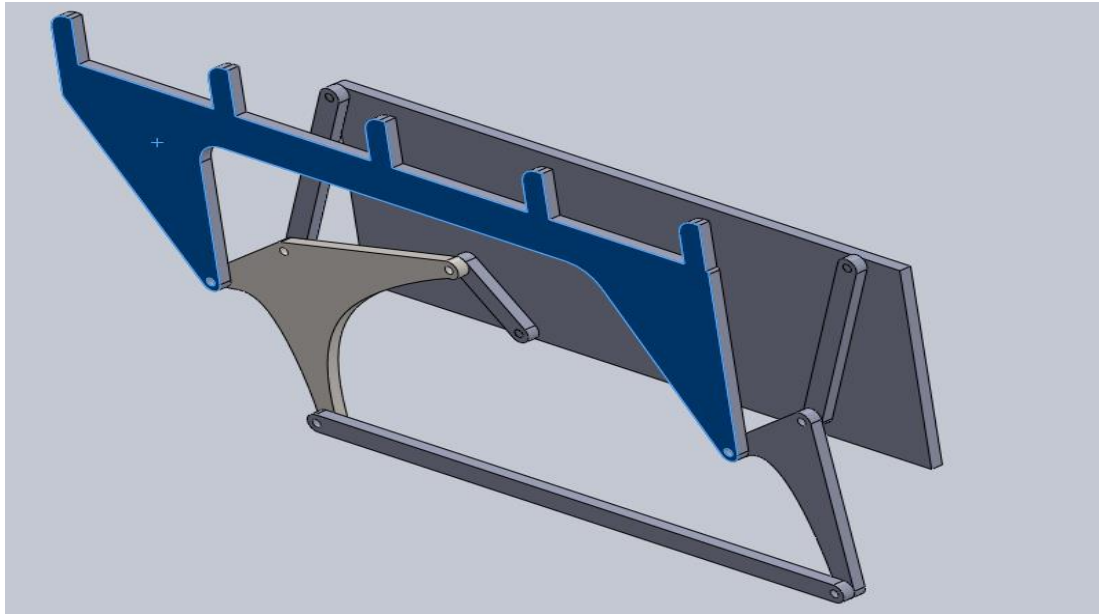


Fig.5 Solid works model showing output link.

Here output link EH (carrier) which is shown by blue colour in fig.5 of solid work model it applies forces on boxes to transfer from one location to other location i.e., it pushes the box in left direction for first half revolution of crank (ccw) while simulating.

2: KINEMATIC SIMULATION

2.1: Output component

Model is simulated for input of angular velocity *19.098 RPM*.

Link EH is the output link. For kinematic simulation of output motion, motion of point(E) as P shown in Assignment 1 Fig.8 is analysed.

Also, the coordinates system along with sign convention of expected and simulation are consistent.

Link EH is the output link. For kinematic simulation of output motion, motion of point P as shown in Assignment 1 Fig.8 is analysed. Model is simulated for input of angular velocity *19.098 RPM*.

Minor Changes

1. Angles on the predicted graph vary from 0 to 2π , however in the simulation performed by Solid Works, angles range from $-\pi$ to π .
2. Here all the simulated graph is plotted for one complete revolution of crank here crank speed is 19.058 RPM means time for one cycle is approximately 3 sec

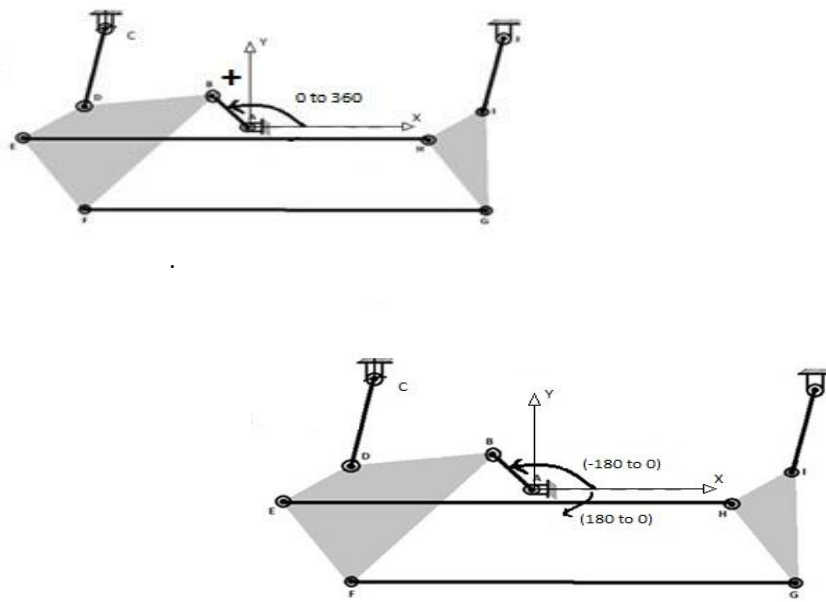


Fig.6 Expected Coordinate System and sign convention

Plot of angular displacement of link AB w.r.t time for one revolution of crank is given below .

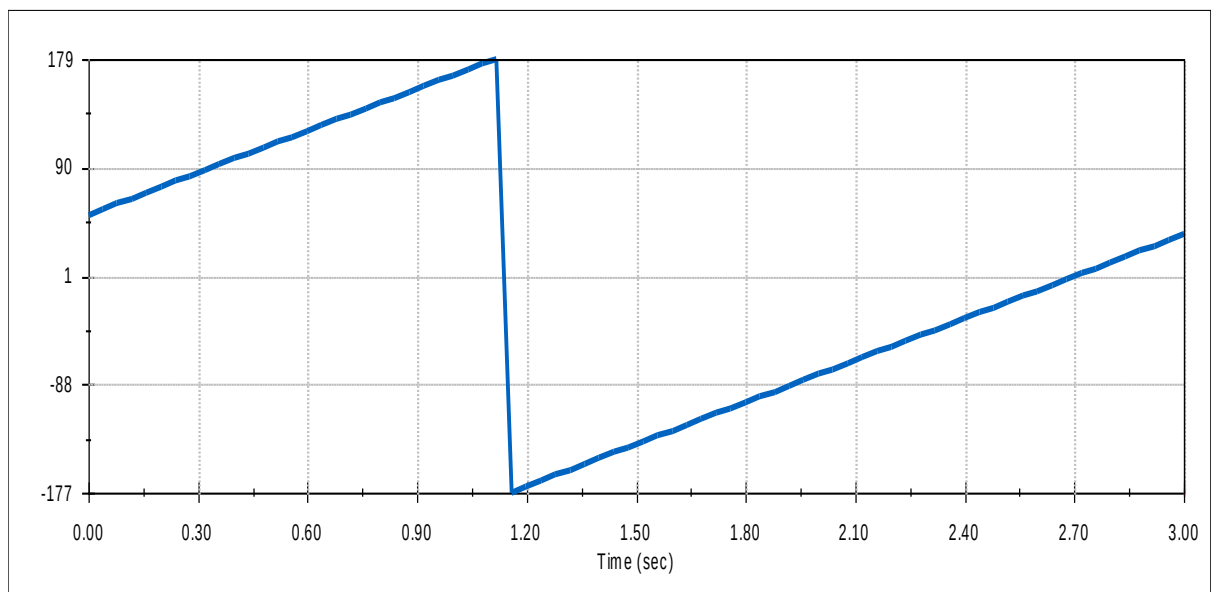
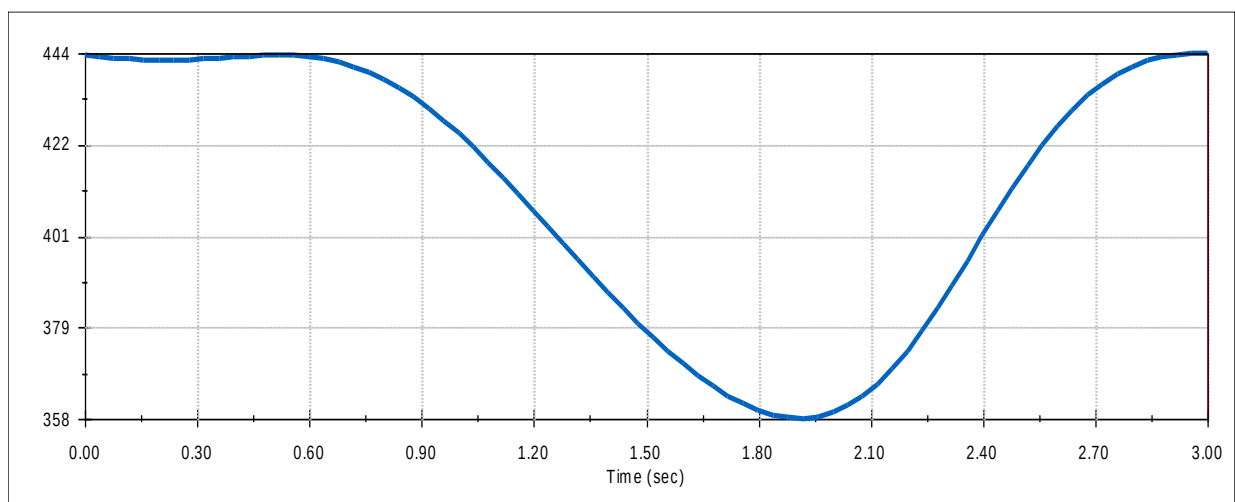
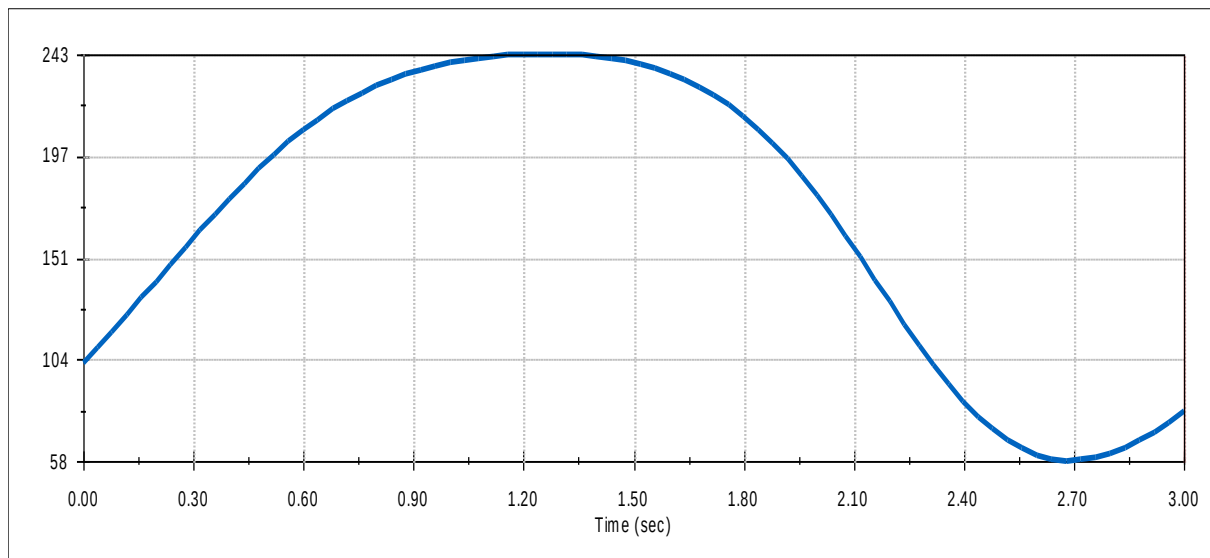
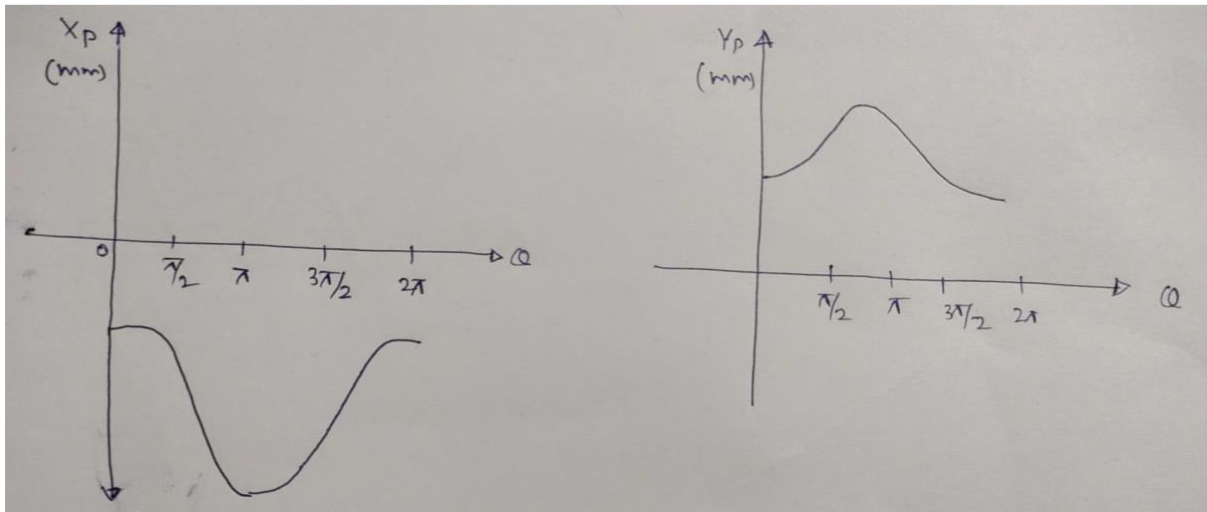


Fig.7 Angular displacement of link AB Vs time

2.1.1: Simulated and expected displacement profile of the output component



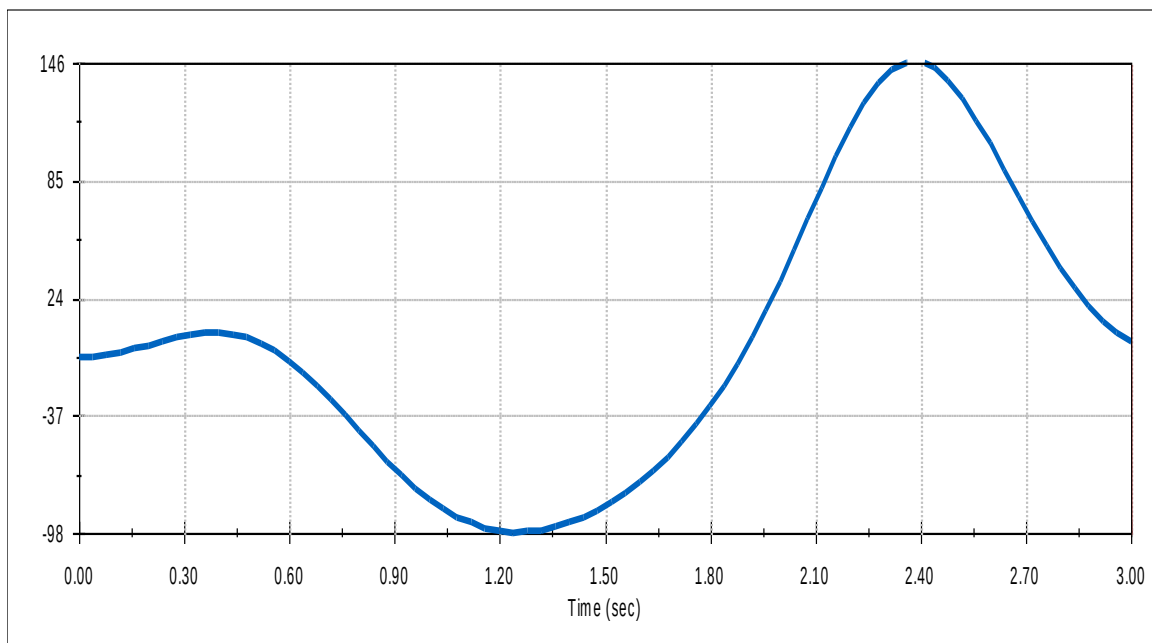
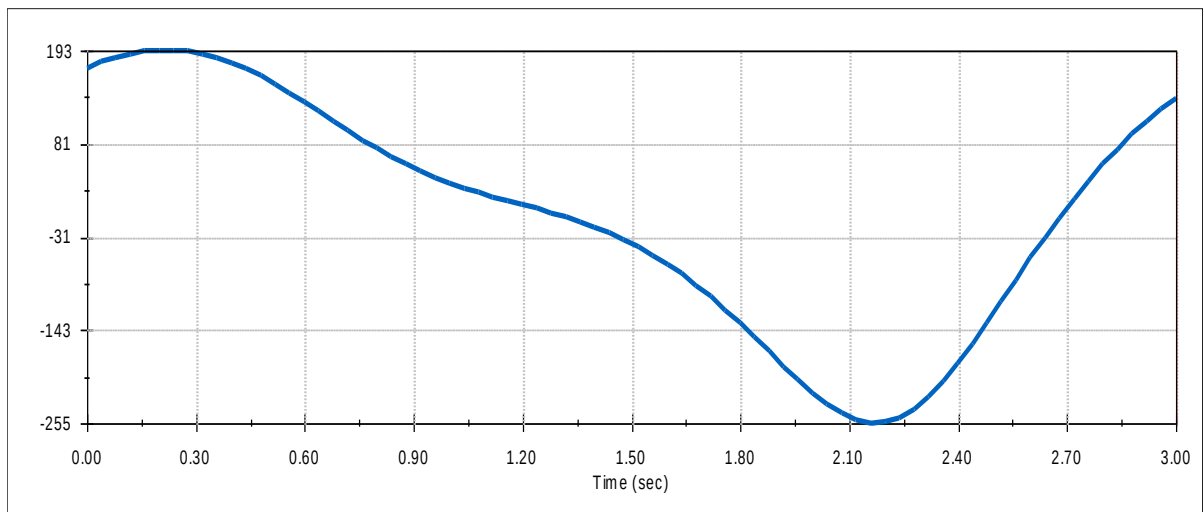
Expected displacement profile



Due to sign convection I used for plotting expected graph and sign convention that solid work has taken to plot simulated graph is different hence graph of displacement of simulated(up) and theoretical(down) is showing .

Fig.8 (Top) X – comp. of displacement and (Bottom) Y – comp. of displacement

2.1.2 :Simulated and expected velocity profile of the output component



Expected velocity profile

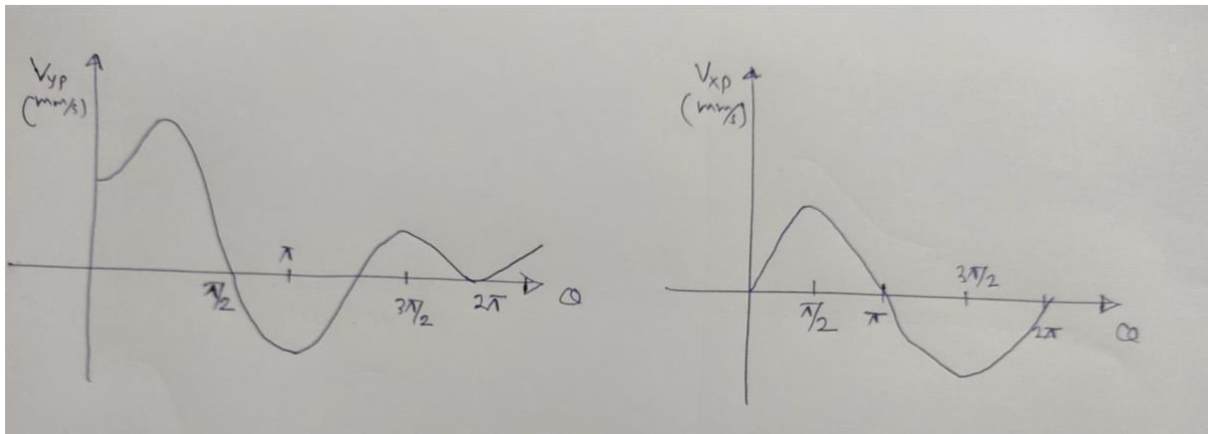
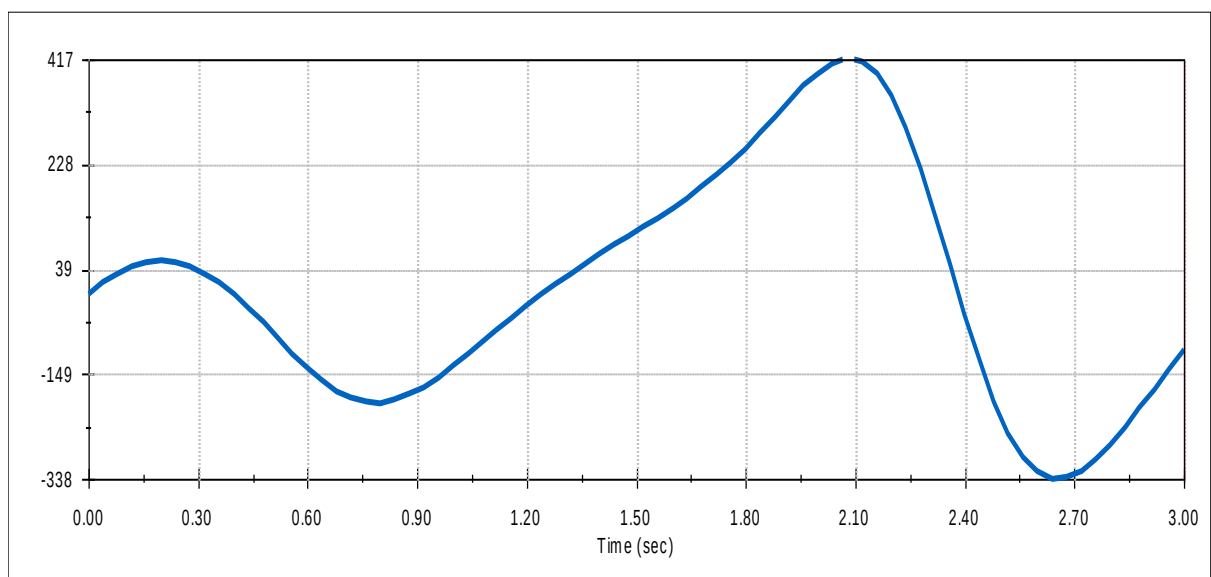
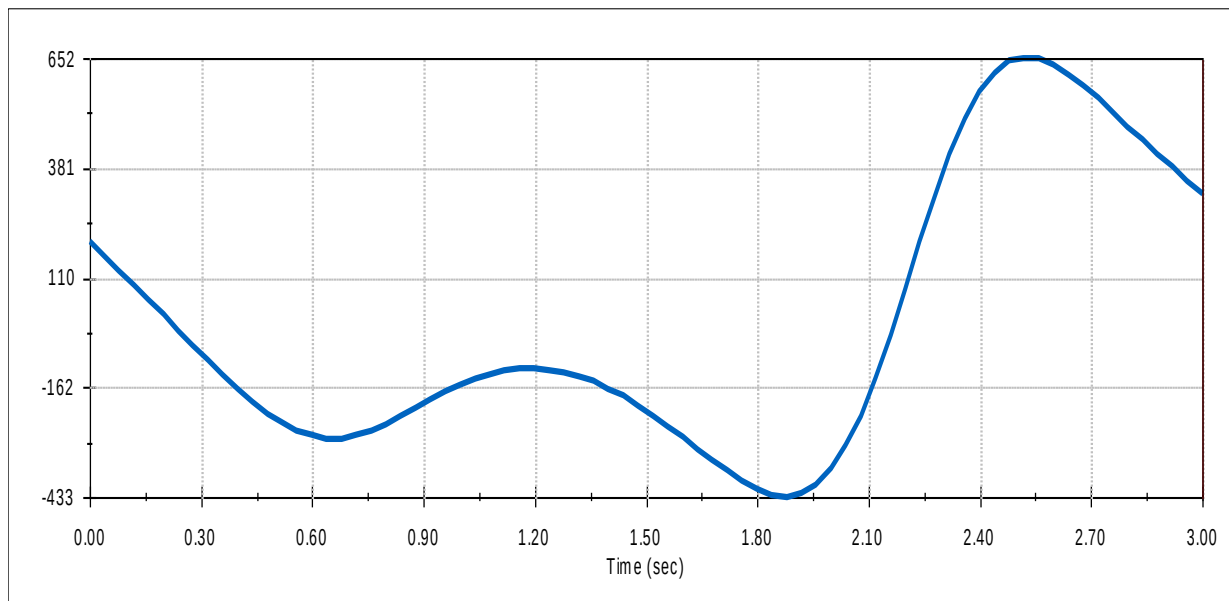


Fig.9 (Top) X – comp. of velocity and (Bottom) Y – comp. of velocity

2.1.3: Simulated and expected acceleration profile of the output component



Expected acceleration profile

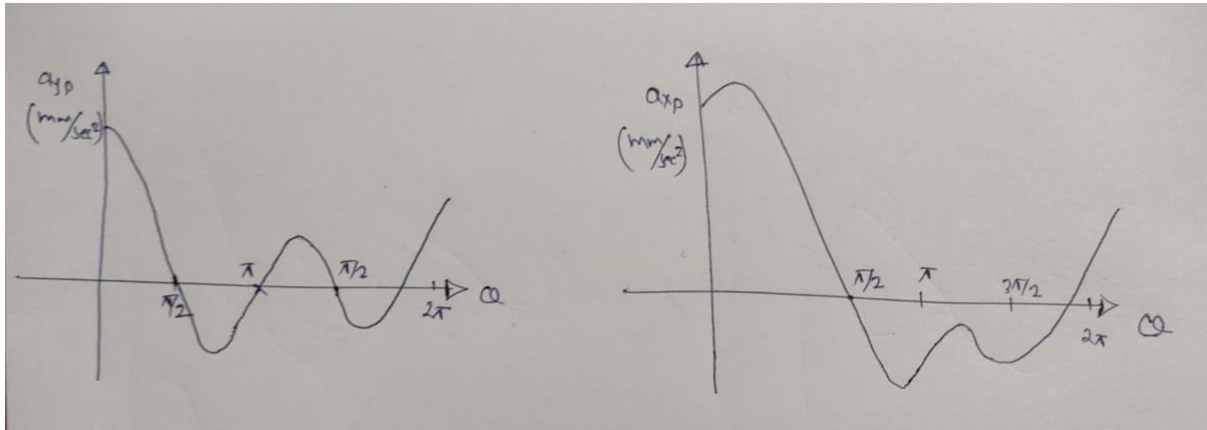


Fig.10 (Top) X – comp. of acceleration and (Bottom) Y – comp. of acceleration

2.1.4 Comments

Displacement (Fig.8): Since the dimensions are changes it can be seen change in magnitude of displacement and time corresponding to magnitude get changed error another reason also that expected graph is hand drawn and not to scale and not only sign convention but also a coordinate system and simulated initial condition are not same. as crank rotates point E in output link follow the same pattern but only difference is the sign because of sign convention we choose is different than expected one due to this there is difference in sign is observed but follows the same pattern as in expected one .

Velocity (fig 9): Expected and simulated both component of velocity have similar graph except that it doesn't coincide because of being hand drawn and its working is similar to crank rocker mechanism in quick return mechanism the simulated one attain maximum negative value around 2.2 sec but in expected one it attains maximum negative value during one cycle at 1.57 sec for x _ velocity and for y_ velocity at 1.2 sec and for expected one its around 2.3 sec. and its magnitude is also differed because of change in dimension error are amplified due to error in linkage dimension. And the crank rotates counter clockwise from 0 to 180 due to this output link having teeth apply forces on boxes and from 180 to 360 degree rotation of crank the carrier dips and come to initial position.

Acceleration (fig 10): Expected and simulated both component of acceleration have highest mismatch as compared to displacement and velocity one of the reason may be we are not taking account of angular velocity of carrier link EH and other reason will be expected graph is hand drawn and not scaled one and error due to link dimension but it follow the similar pattern as expected only magnitude and time shift is seen due to dimension of link chosen is different .these are the thing which will build up error

2.2: SENSITIVITY ANALYSIS

Now we will increase the length of two link by 10 mm and see how much its is sensitive and plot the graph of kinematic parameter of output link .

Case 1: Crank Link (AB): 91.95mm

Modified length (AB): $91.95 + 10 = 101.95\text{mm}$

Case 2: Rocker Link (IJ): 234.75mm

Modified length (IJ): $234.75 + 10 = 244.75\text{mm}$

2.2.1 :Sensitivity Analysis due to Crank Link AB

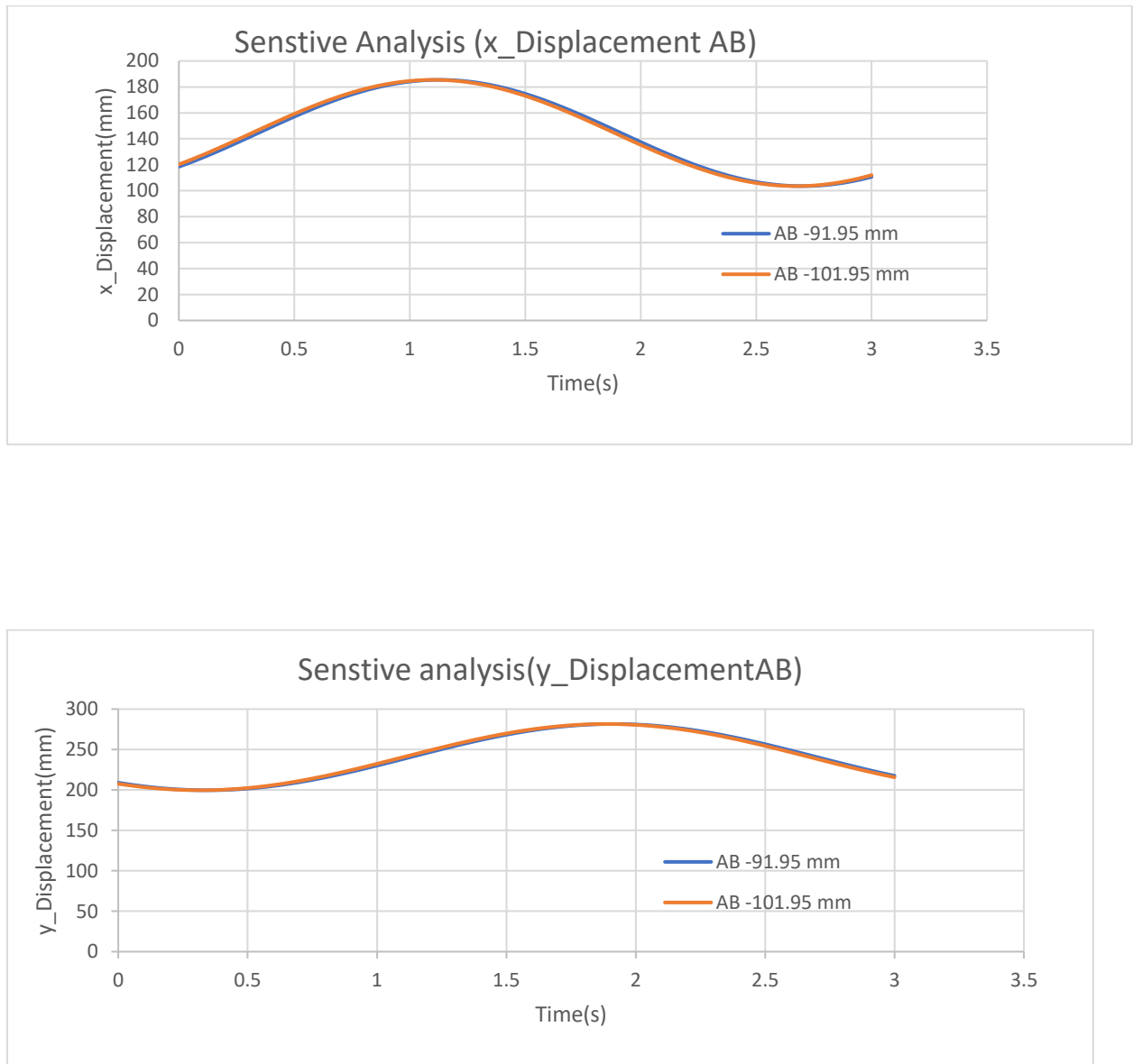


Fig-11 (Top) X - comp of displacement and (Bottom) Y – comp of displacement

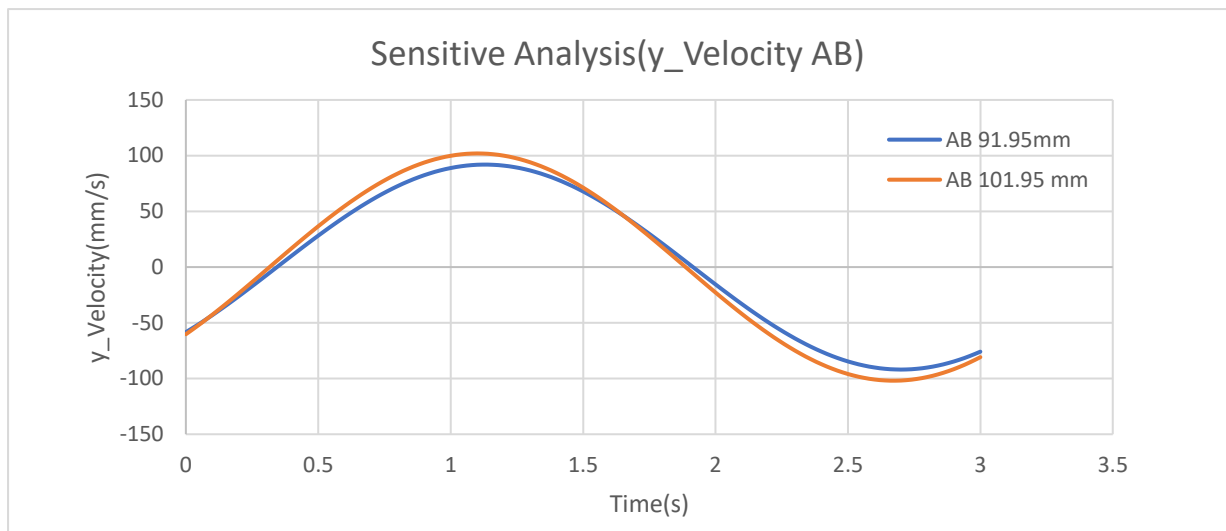
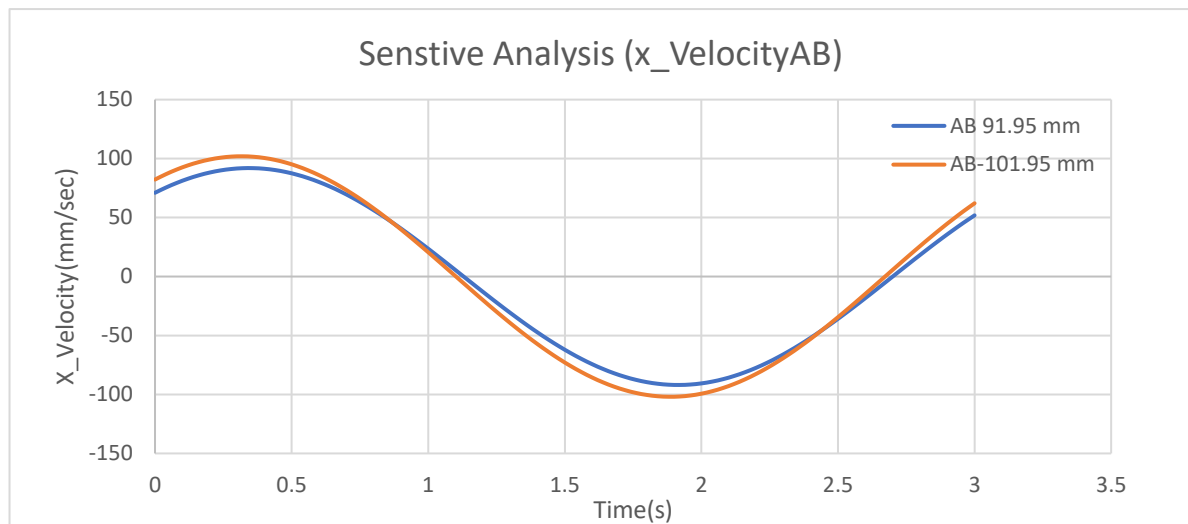


Fig.12 (Top) X - comp of Velocity and (Bottom) Y – comp of Velocity

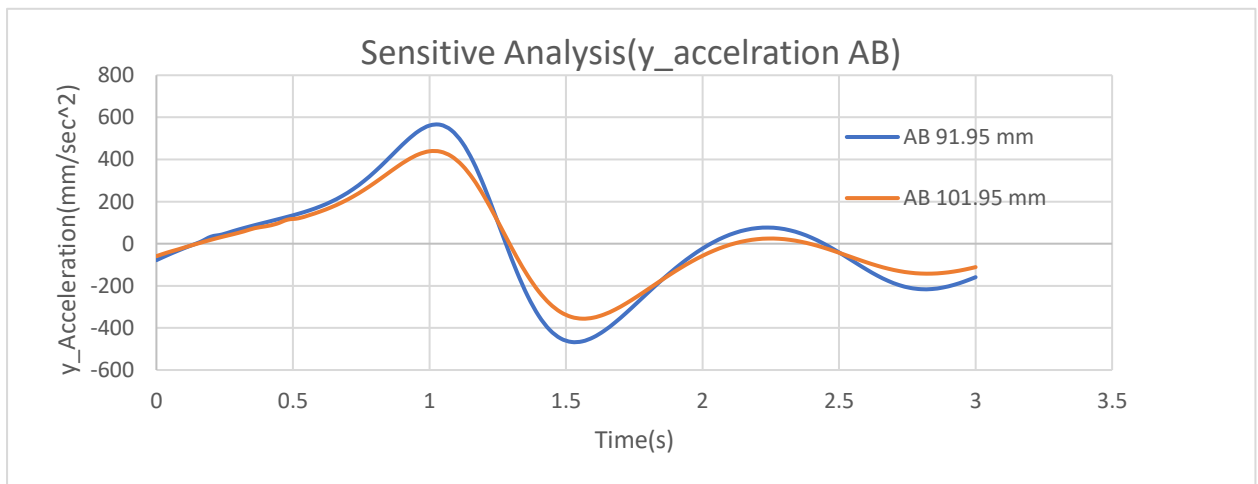
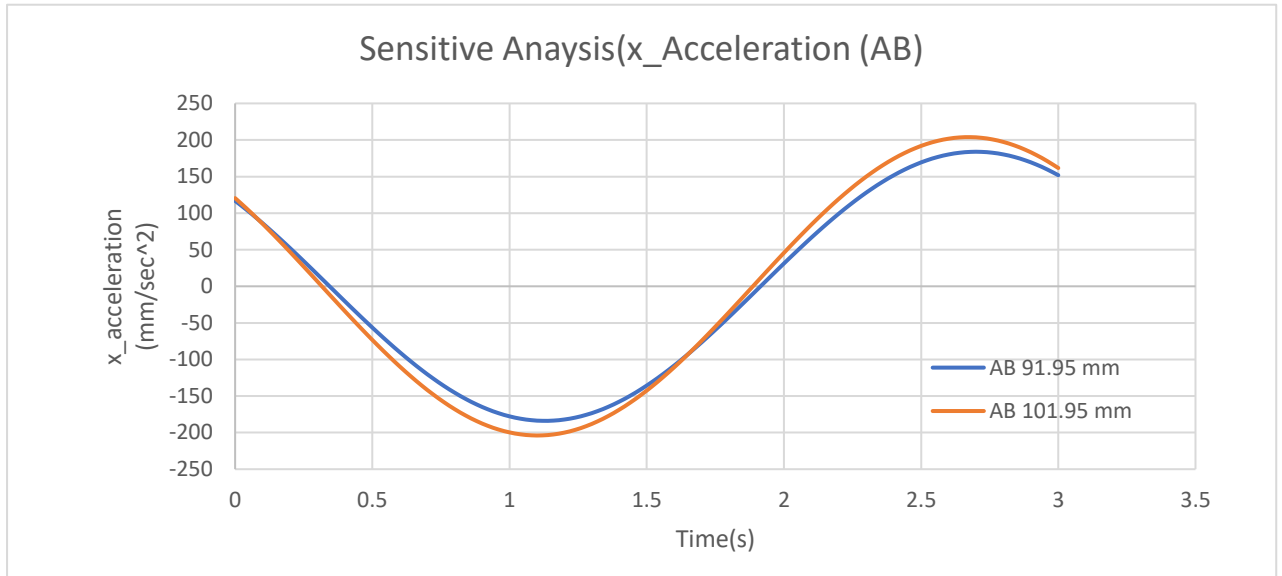


Fig.13 (Top) X - comp of Acceleration and (Bottom) Y – comp of Acceleration

2.2.1: Sensitivity analysis due to Rocker Link IJ

The variations of Kinematic parameter in output link due to increase of 10 mm length to rocker (IJ) observed and is shown in figures below.

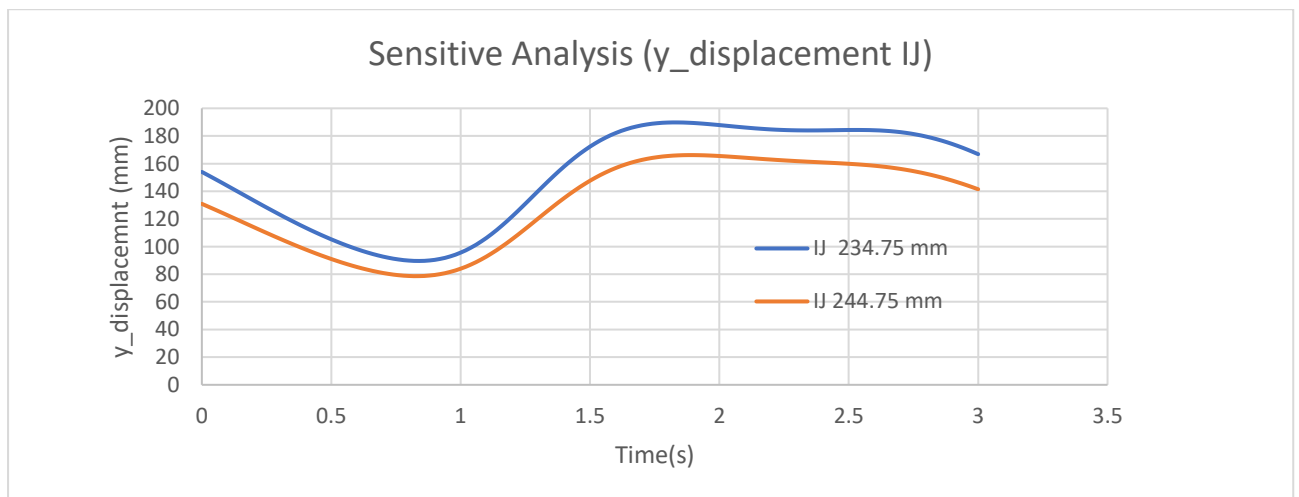
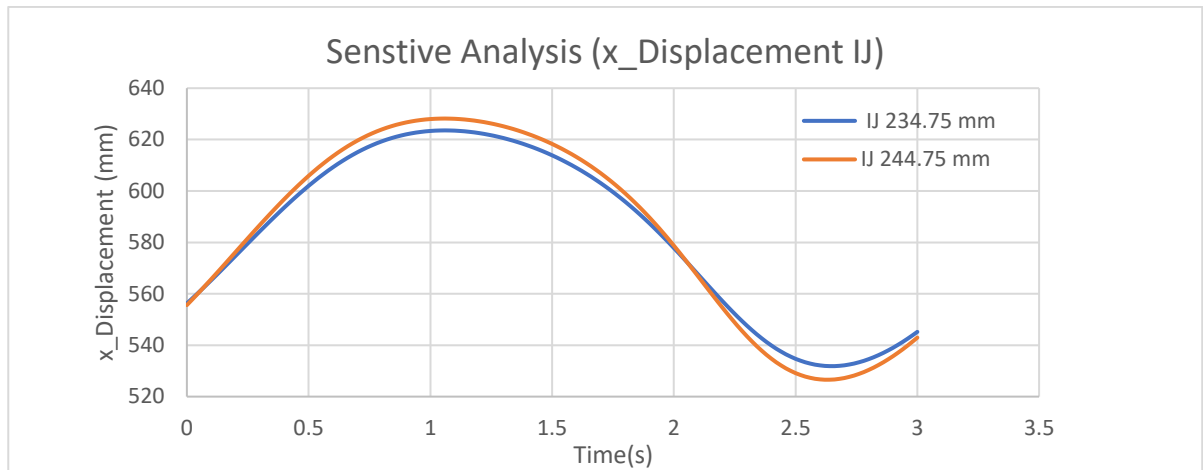


Fig. 14 (Top) X - comp of displacement and (Bottom) Y – comp of displacement

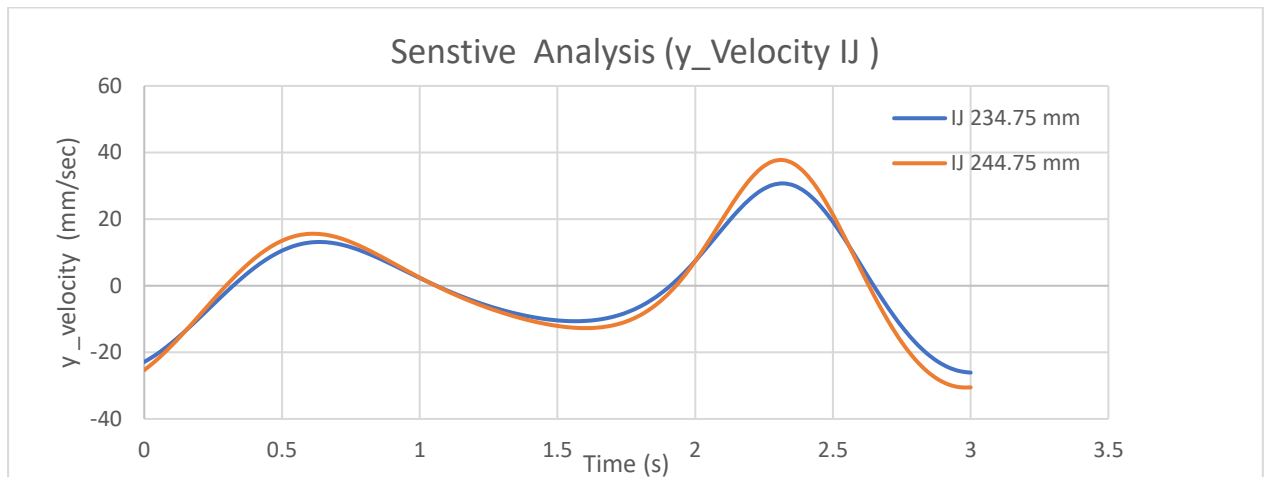
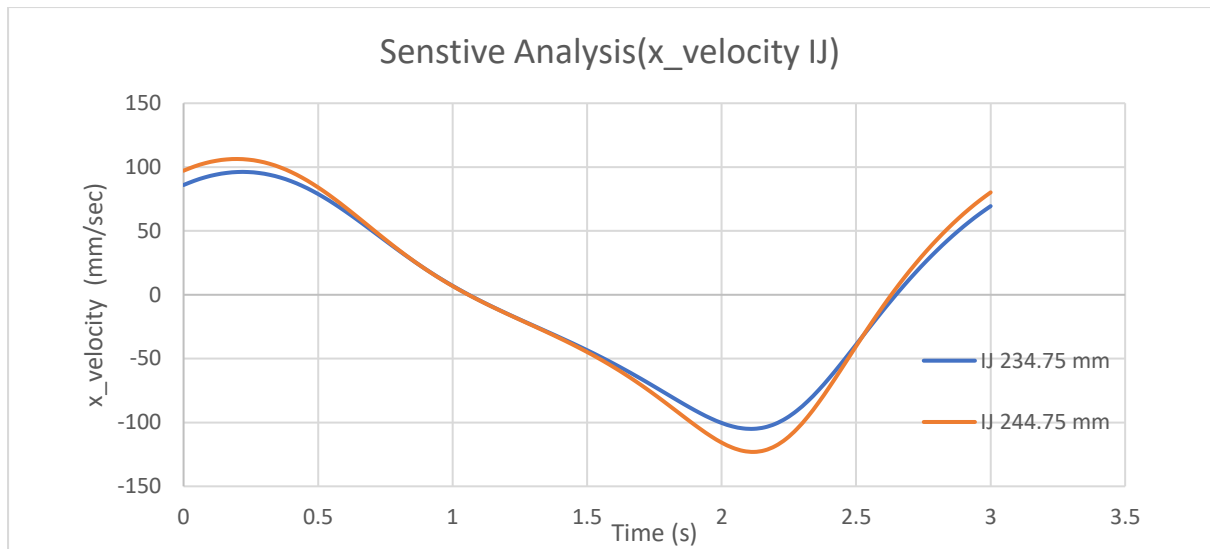


Fig.15 (Top) X - comp of velocity and (Bottom) Y – comp of velocity

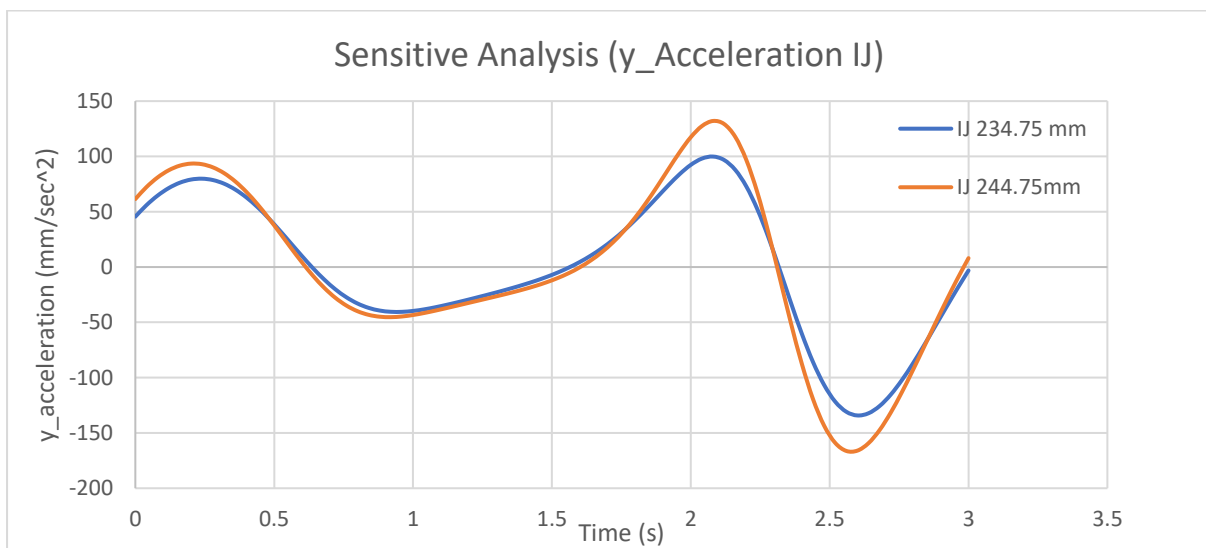
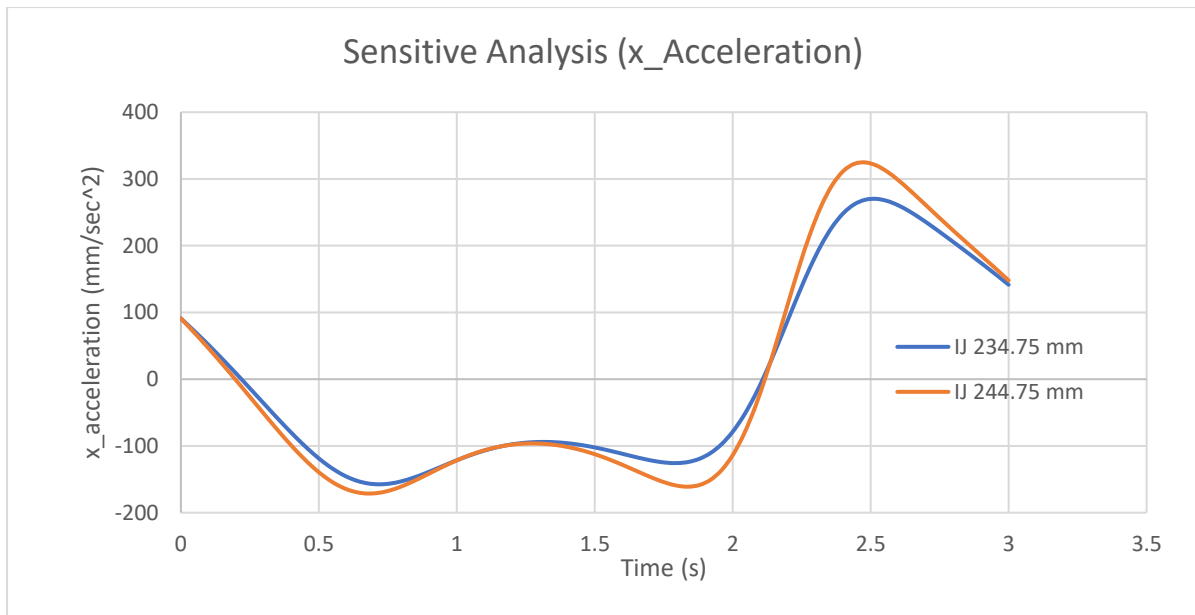


Fig.16 (Top) X - comp of acceleration and (Bottom) Y – comp of acceleration

2.3: ANALYSIS OF JOINT CLEARANCE

Simulation is done for crank at normal speed (19.098 RPM) and at high speed (114.588 RPM) for Time duration of 3 sec is taken for simulation at both speeds.

2mm of diametrical clearance is given at J and the variation of kinematic parameter is observed due to clearance.

2.3.1 Normal Speed and High Speed

Each graph contains three curves one is same as simulated graph (Blue, Sec. 2.1), other is for clearance at normal speed (Red) and the last one is for clearance at high speed (Green)

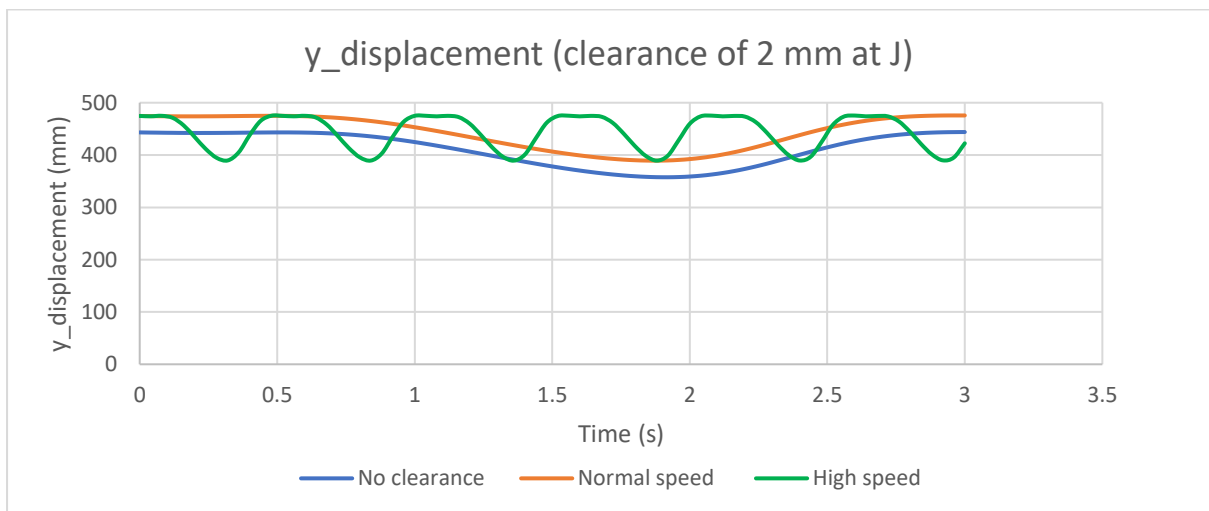
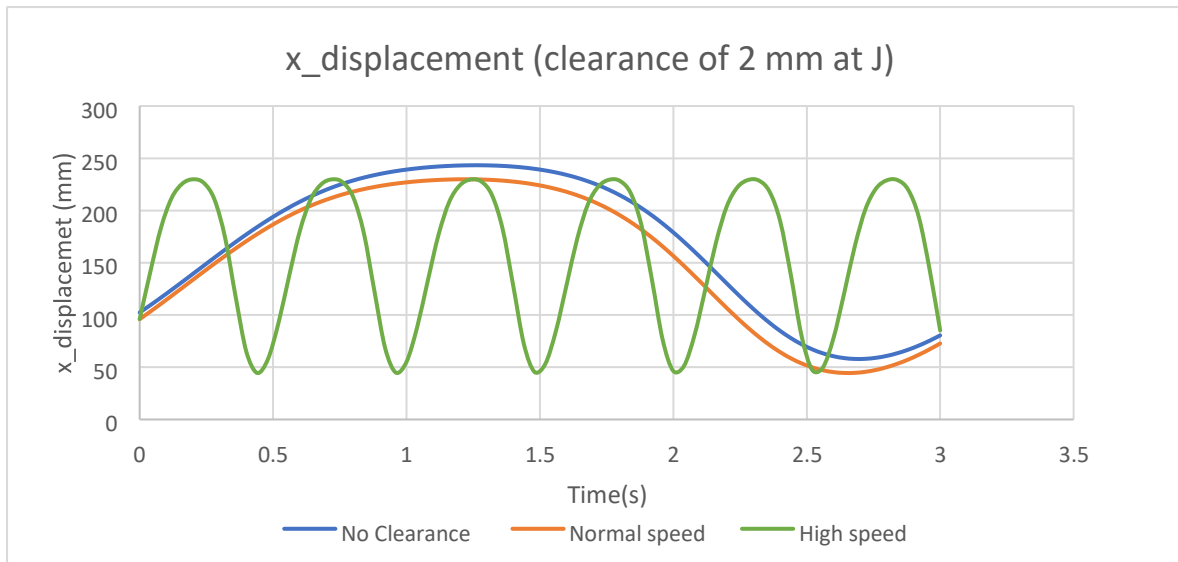


Fig.17 (Top) X - comp of displacement and (Bottom) Y – comp of displacement

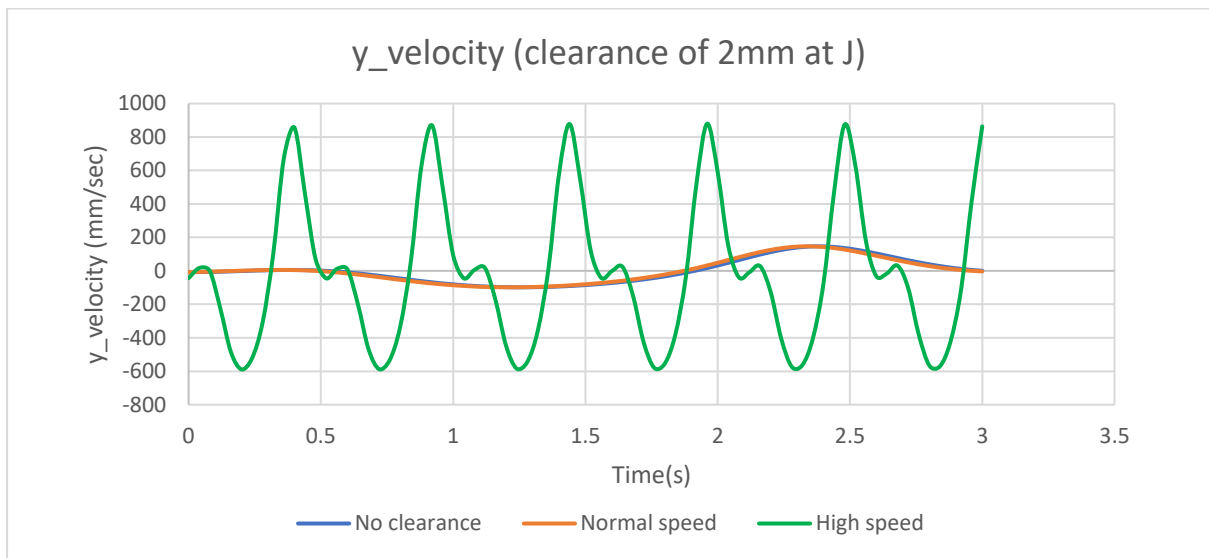
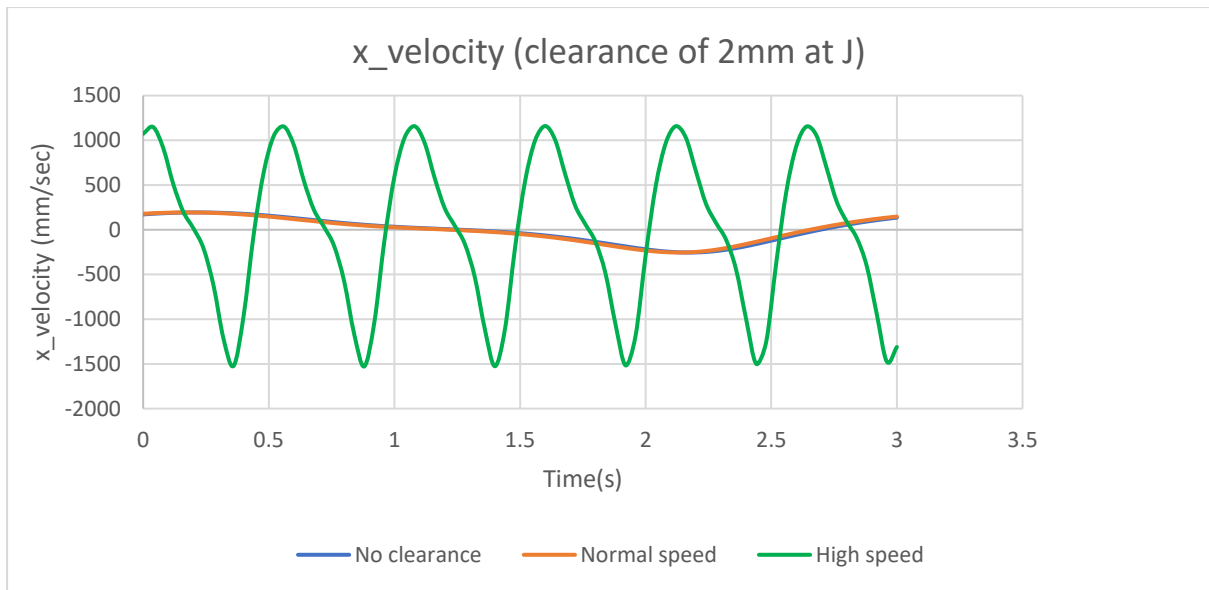


Fig.18 (Top) X - comp of velocity and (Bottom) Y – comp of velocity

When we plot all three graph and observe that blue line is not visible and red line follows blue line approximately.

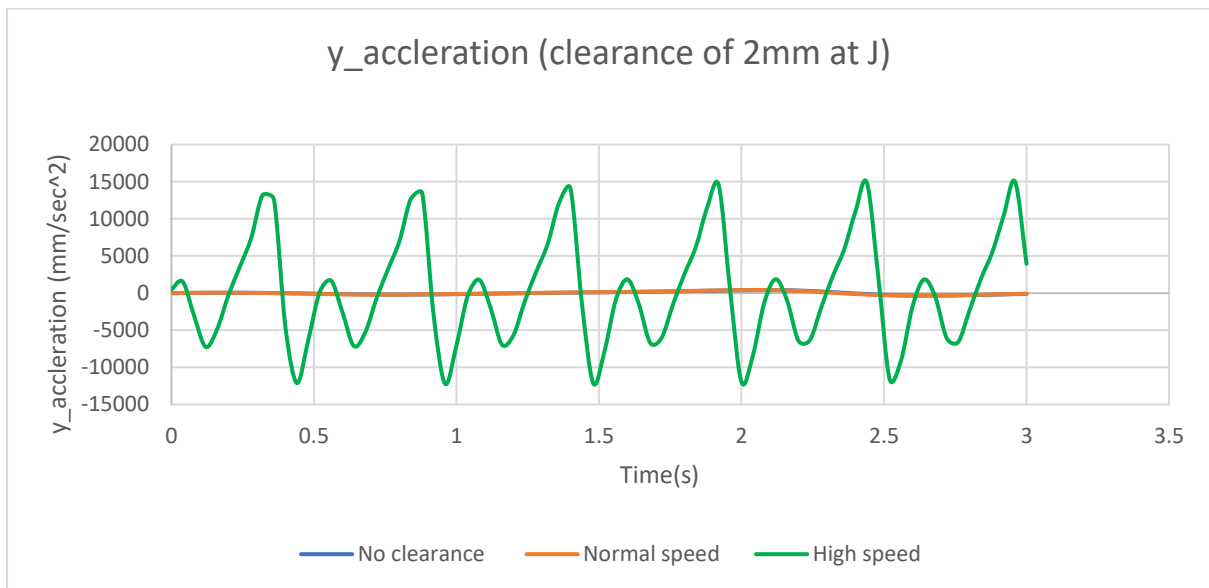
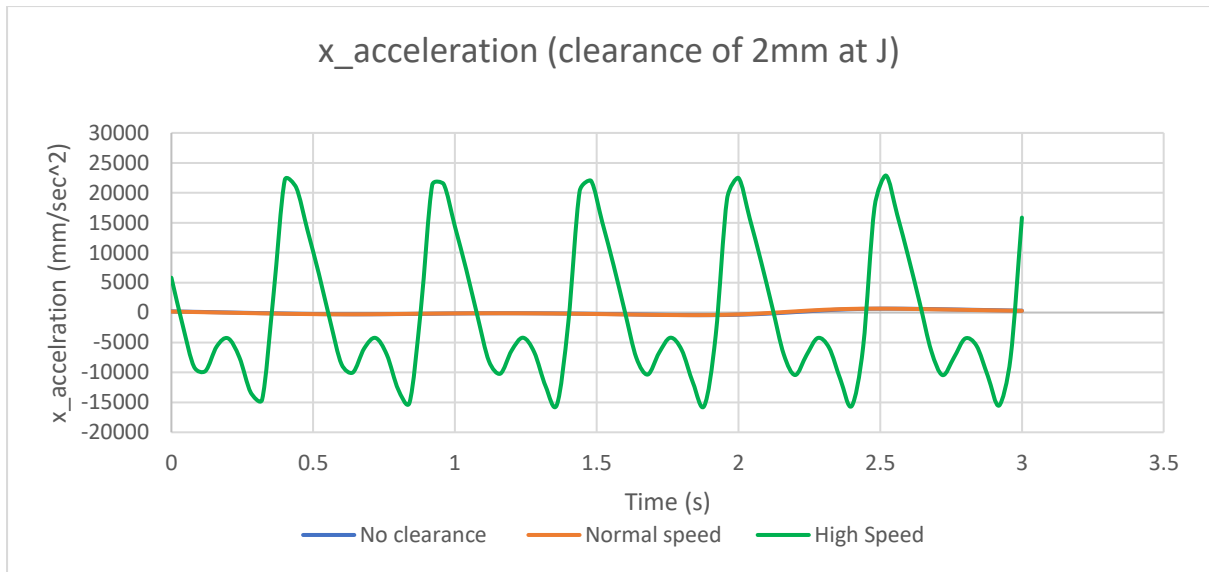


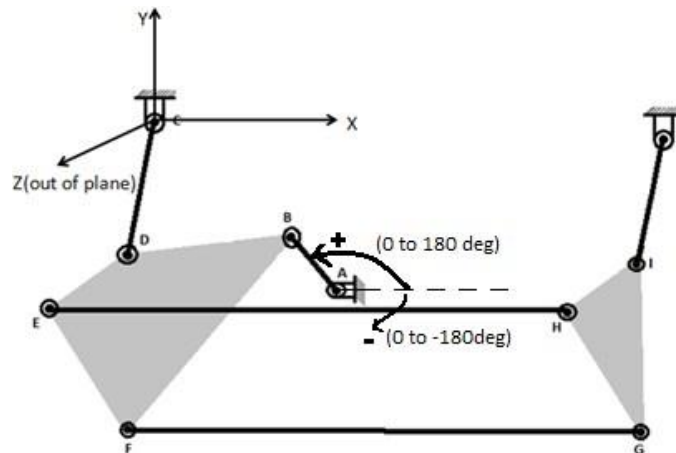
Fig.19 (Top) X - comp of acceleration and (Bottom) Y – comp of acceleration

Due to large gap in amplitude blue line is not visible. All line follows same as acceleration graphs in section 2.1.3. with sudden jerks (for green lines) at different time intervals.

3: VALIDATION OF KINEMATIC SIMULATION

3.1: Theoretical evaluation of kinematic

For analytical analysis, three kinematic parameters i.e. displacement, velocity and acceleration of point E is evaluated for four value of crank AB's input angle {0 deg, (90) deg, 180 deg and 270 deg}.



To find displacement velocity and acceleration of output link at E different position of crank we draw ,crank at 0, 90 ,180 ,270 degree and find coordinate of pt A,B,D,E,C from solid works and these displacement will become input for finding velocity and acceleration of output link.

Coordinates of points B,D,E calculated graphically and shown in table below .

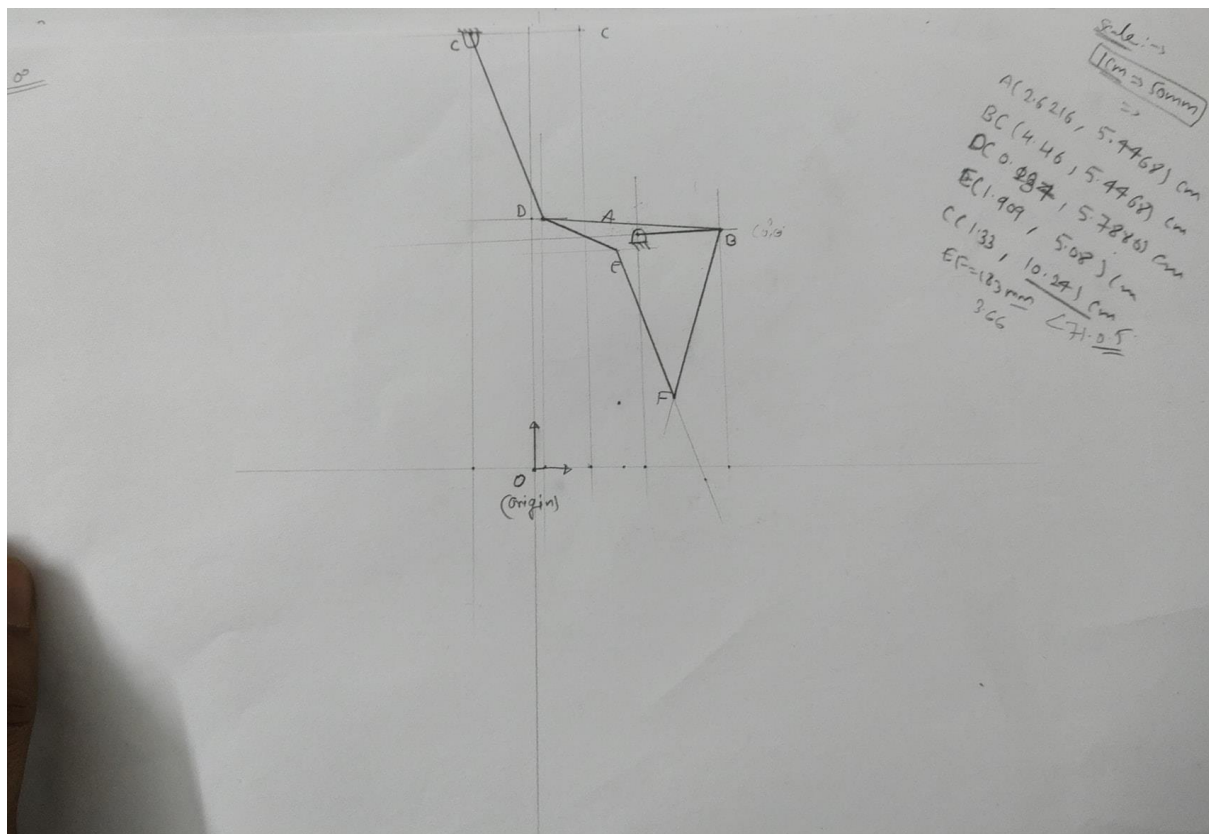
Input Angle (degree)	B (x, y) mm	D (x, y) mm	E (x, y) mm
0	(233.03,272.34)	(6.7,289.43)	(-95.47,254.43)
90	(131.08,364.29)	(-67.92,277.62)	(-141.54,198.63)
180	(39.13,272.34)	(-175.52,304.22)	(-279.85,276.53)
270	(131.05,180.39)	(-62.92,277.62)	(-170.75,283.64)

Point A and C are fixed point A Coordinates (131.08,272.34) point D coordinates (-66.92,512.34) and coordinates of other points are tabulated above .

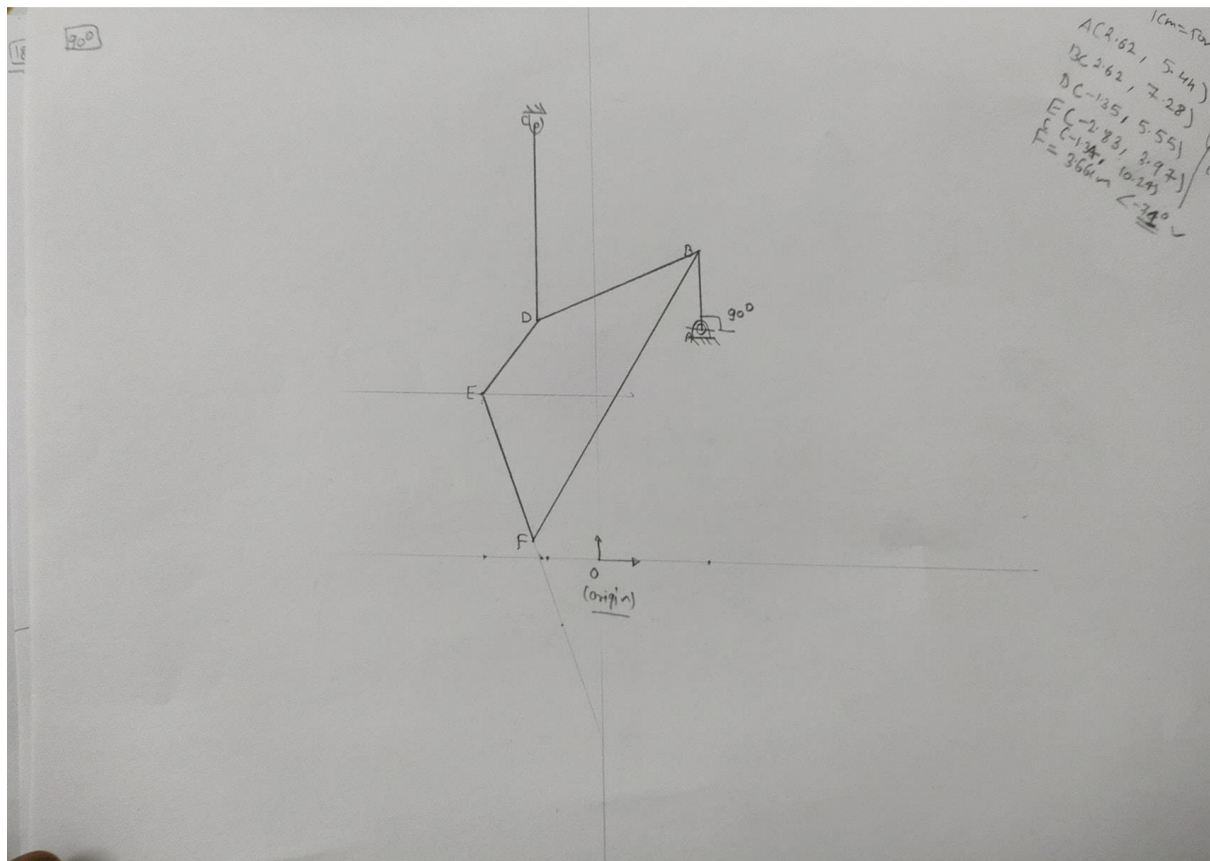
3.1.1: Calculating Displacement Graphically

To calculate velocity and acceleration of point E at different position of crank we need rotate the crank at different position and obtained other points coordinates (i.e B,D,E) from solid work and its orientation in different position is drawn graphically and these point now become input parameter for velocity and acceleration.

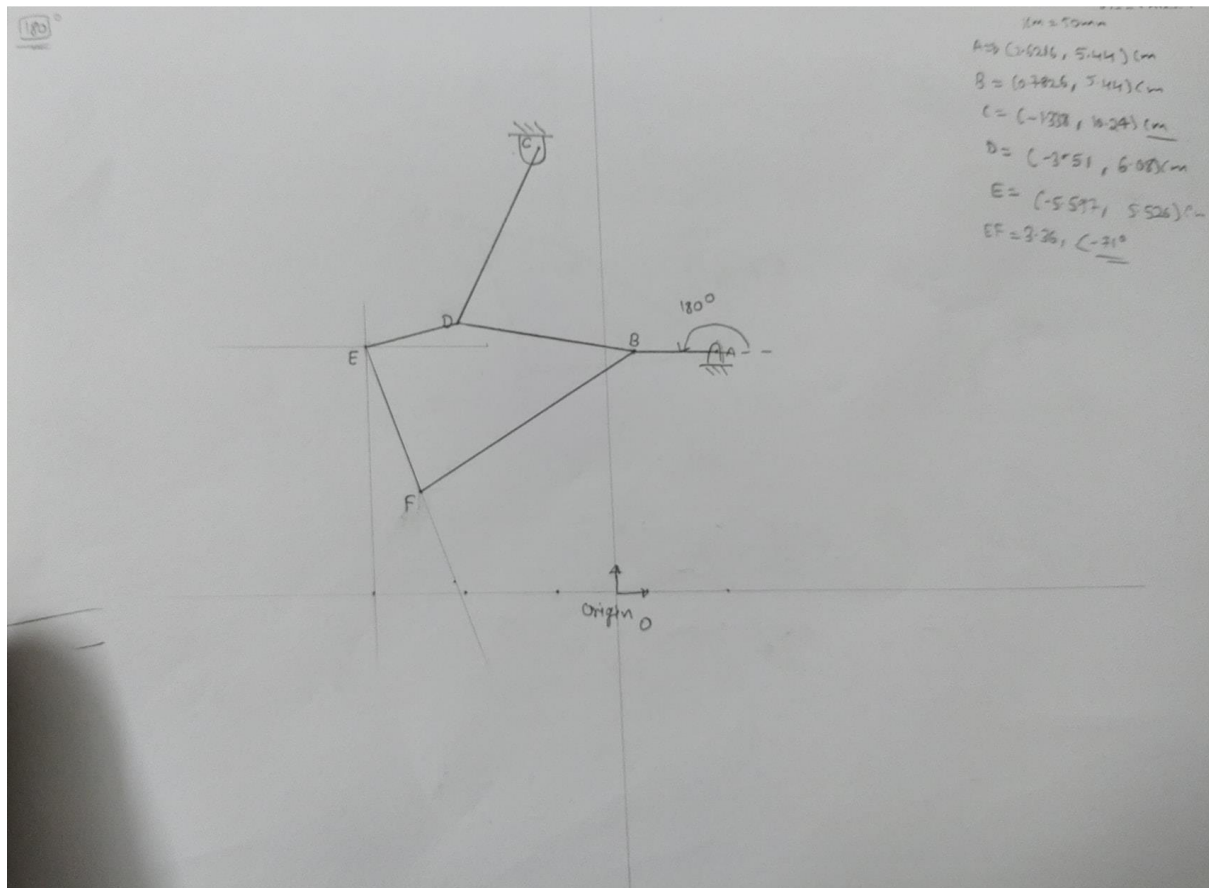
0 Degree crank position



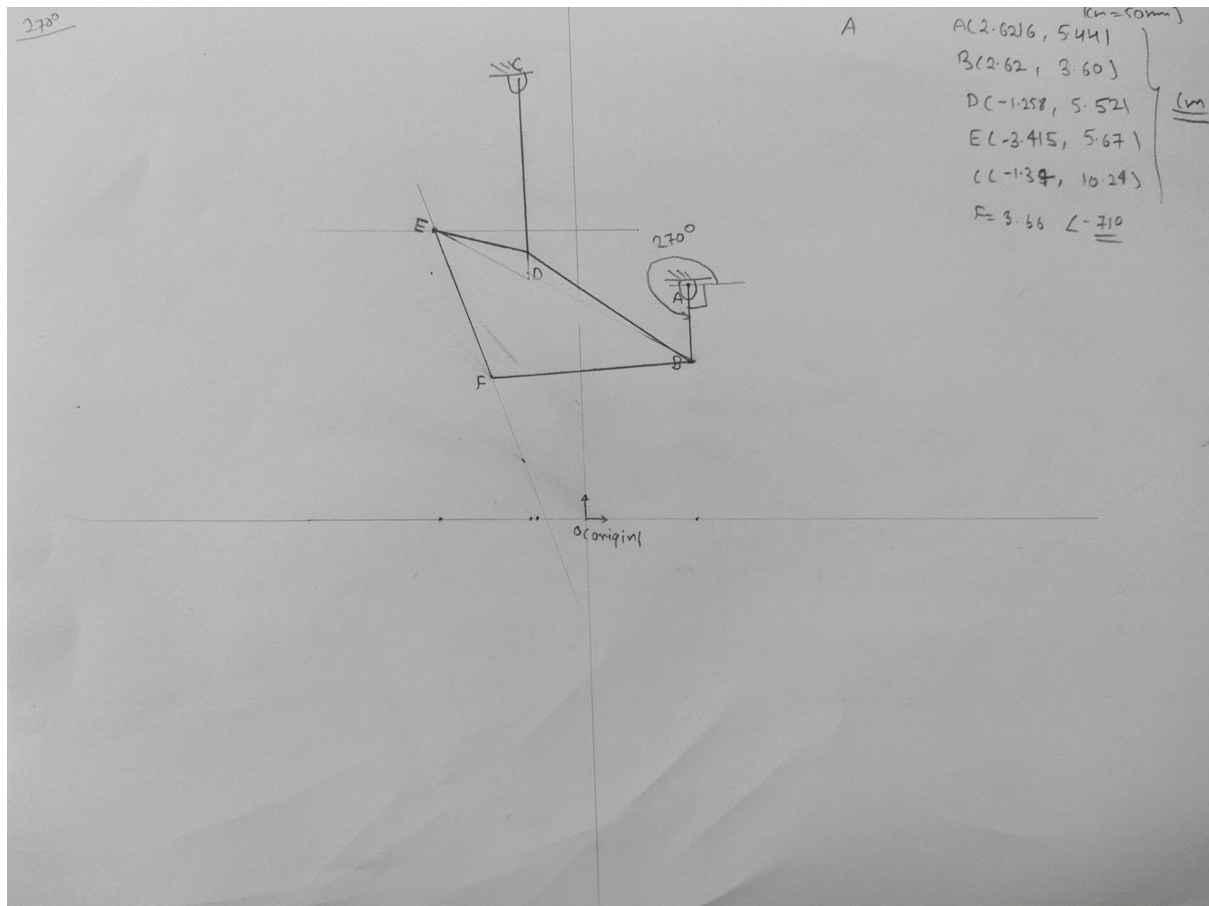
90- degree crank position



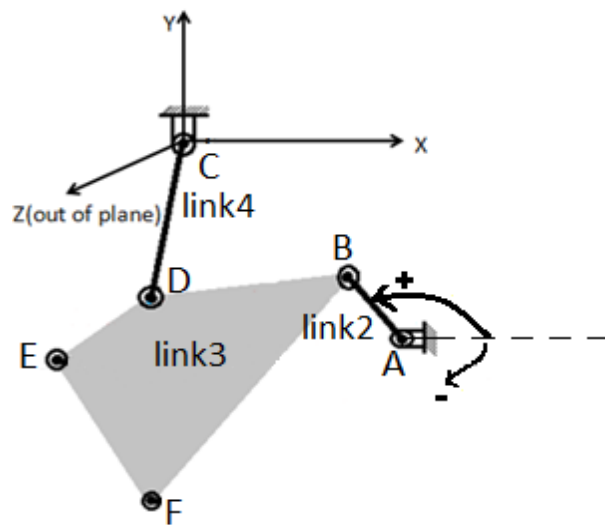
180-degree crank position



270 – degree crank position



3.1.2 Calculating velocity and acceleration analytically and coding



To find velocity and acceleration of point E we are using below analytical equation and these equations are derived below. These equations need displacement values which are obtained from graphical method position of all points at different position of crank.

Velocity analysis : -

$$V_B = \cancel{V_A} + \omega_2 \times \vec{r}_{AB} = 0 + \omega_2 \times \vec{r}_{AB}$$

$$V_D = V_B + \omega_3 \times \vec{r}_{BD} = \omega_2 \times \vec{r}_{AB} + \omega_3 \times \vec{r}_{BD} \rightarrow \textcircled{1}$$

Also

$$V_D = V_C + \omega_4 \times \vec{r}_{CD} \quad \boxed{V_C = 0}$$

$$V_D = \omega_4 \times \vec{r}_{CD} \rightarrow \textcircled{2}$$

equating $\textcircled{1}$ & $\textcircled{2}$

$$\omega_4 \hat{k} \times (r_{CD} \hat{i} + r_{CD} \hat{j}) = \omega_2 \hat{k} \times (r_{AB} \hat{i} + r_{AB} \hat{j}) + \omega_3 \hat{k} \times (r_{BD} \hat{i} + r_{BD} \hat{j})$$

$$\omega_4 r_{CD} \hat{j} - \omega_4 r_{CD} \hat{i} = \omega_2 r_{AB} \hat{j} - \omega_2 r_{AB} \hat{i} + \omega_3 r_{BD} \hat{i}$$

$$\omega_4 r_{CD} \hat{j} - \omega_3 r_{BD} \hat{j} = \omega_2 r_{AB} \hat{j}$$

$$\omega_4 r_{CD} \hat{i} - \omega_3 r_{BD} \hat{i} = \omega_2 r_{AB} \hat{i}$$

$$\begin{bmatrix} r_{CD} \hat{i} & -r_{BD} \hat{i} \\ r_{CD} \hat{j} & -r_{BD} \hat{j} \end{bmatrix} \begin{bmatrix} \omega_4 \\ \omega_3 \end{bmatrix} = \omega_2 \begin{bmatrix} r_{AB} \hat{i} \\ r_{AB} \hat{j} \end{bmatrix}$$

$$a_B = a_A + \alpha_2 \times r_{AB} - \omega_2^2 r_{AB} = -\omega_2^2 r_{AB}$$

$$a_D = a_B + \alpha_3 \times r_{BD} - \omega_3^2 r_{BD} \rightarrow (3)$$

$$a_D = a_C + \alpha_4 \times r_{CD} - \omega_4^2 r_{CD} \rightarrow (4)$$

equating (3) & (4)

$$\begin{aligned} \alpha_4 \times (r_{CD} \hat{i} + r_{CD} \hat{j}) - \omega_4^2 (r_{CD} \hat{i} + r_{CD} \hat{j}) \\ = -\omega_2^2 (r_{AB} \hat{i} + r_{AB} \hat{j}) + \alpha_3 \times (r_{BD} \hat{i} + r_{BD} \hat{j}) \\ - \omega_3^2 (r_{BD} \hat{i} + r_{BD} \hat{j}) \end{aligned}$$

Equating i & j component

$$\begin{bmatrix} -r_{CD} \hat{j} & r_{BD} \hat{j} \\ r_{CD} \hat{i} & -r_{BD} \hat{i} \end{bmatrix} \begin{bmatrix} \alpha_4 \\ \alpha_3 \end{bmatrix} = \omega_4^2 \begin{bmatrix} r_{CD} \hat{i} \\ r_{CD} \hat{j} \end{bmatrix} - \omega_2^2 \begin{bmatrix} r_{AB} \hat{i} \\ r_{AB} \hat{j} \end{bmatrix} - \omega_3^2 \begin{bmatrix} r_{BD} \hat{i} \\ r_{BD} \hat{j} \end{bmatrix}$$

Velocity & acceleration of pt E

$$\begin{aligned} v_E &= v_B + \omega_3 \times \vec{r}_{BE} = (\omega_2 \times \vec{r}_{AB}) + (\omega_3 \times \vec{r}_{BE}) \\ &= -(\omega_2 r_{AB} \hat{j} + \omega_3 r_{BE} \hat{j}) \hat{i} + (\omega_2 r_{AB} \hat{i} \\ &\quad + \omega_3 r_{BE} \hat{i}) \hat{j} \rightarrow (5) \end{aligned}$$

$$\begin{aligned} a_E &= a_B + \alpha_3 \times \vec{r}_{BE} - \omega_3^2 r_{BE} = -\omega_2^2 r_{AB} \\ &\quad + \alpha_3 \times \vec{r}_{BE} - \omega_3^2 r_{BE} \end{aligned}$$

$$\underline{\underline{a_E}} = (-\omega_2^2 r_{AB} x - \omega_3^2 r_{BE} x - \kappa_3 r_{BE} y) \hat{i}$$

$$+ (-\omega_2^2 r_{AB} y - \omega_3^2 r_{BE} y + \kappa_3 r_{BE} x) \hat{j}$$

→ (6)

Combing equation 1 to 6 and displacement of point E is calculated graphically and these value we given in MATLAB code as input for calculating velocity acceleration of output link at different instant of mechanism .below are screenshot of MATLAB code is attached

```

1      %input at 0 degree
2      a = [131.08;272.34];
3      c = [-66.92; 512.34];
4      b = [223.03;272.34];
5      d = [6.7; 289.43];
6      e = [-95.47;254.43];
7
8      ab = b-a;
9      bd = d-b;
10     cd = d-c;
11     be = e-b;
12     w2 = 2;
13     %velocity
14     a1 = [cd -bd];
15     a2 = w2*ab;
16     z = inv(a1)*a2;
17     w4 = z(1,1);
18     w3 = z(2,1);
19     %acceleration
20     b1 = [-cd(2,1) bd(2,1); cd(1,1) -bd(1,1)];
21     b2 = w4^2 * cd - w2^2 * ab - w3^2 * bd;
22     q = inv(b1)*b2;
23     a4 = q(1,1);
24     a3 = q(2,1);
25
26
27     %output
28     vx = -w2*ab(2,1) - w3*be(2,1)
29     vy = w2*ab(1,1) + w3*be(1,1)
30     ax = -w2^2 * ab(1,1)-w3^2 * be(1,1) - a3*be(2,1)
31     ay = -w2^2 * ab(2,1)-w3^2 * be(2,1) + a3*be(1,1)
32

```

vx = 15.6330

vy = -94.1072

ax = -120.6397

ay = -66.3400

Fig.20 Displacement, Velocity, and Acceleration Analytical values at $\theta = 0$ degree using MATLAB code

```

1      %input at 90 degree
2      a = [131.08;272.34];
3      c = [-66.92; 512.34];
4      b = [131.08;364.29];
5      d = [-67.85; 277.59];
6      e = [-141.54;198.63];
7
8      ab = b-a;
9      bd = d-b;
10     cd = d-c;
11     be = e-b;
12     w2 = 2;
13     %velocity
14     a1 = [cd -bd];
15     a2 = w2*ab;
16     z = inv(a1)*a2;
17     w4 = z(1,1);
18     w3 = z(2,1);
19     %acceleration
20     b1 = [-cd(2,1) bd(2,1); cd(1,1) -bd(1,1)];
21     b2 = w4^2 * cd - w2^2 * ab - w3^2 * bd;
22     q = inv(b1)*b2;
23     a4 = q(1,1);
24     a3 = q(2,1);
25
26
27     %output
28     vx = -w2*ab(2,1) - w3*be(2,1)
29     vy = w2*ab(1,1) + w3*be(1,1)
30     ax = -w2^2 * ab(1,1)-w3^2 * be(1,1) - a3*be(2,1)
31     ay = -w2^2 * ab(2,1)-w3^2 * be(2,1) + a3*be(1,1)
32

```

vx = -184.5078
vy = 1.0002
ax = -427.4087
ay = 335.5775

Fig.21 Displacement, Velocity, and Acceleration Analytical values at $\theta = 90$ degree using MATLAB code

```

1      %input at 180 degree
2      a = [131.08;272.34];
3      c = [-66.92; 512.34];
4      b = [39.13;272.34];
5      d = [-175.52; 304.22];
6      e = [-279.85;276.3];
7
8      ab = b-a;
9      bd = d-b;
10     cd = d-c;
11     be = e-b;
12     w2 = 2;
13     %velocity
14     a1 = [cd -bd];
15     a2 = w2*ab;
16     z = inv(a1)*a2;
17     w4 = z(1,1);
18     w3 = z(2,1);
19     %acceleration
20     b1 = [-cd(2,1) bd(2,1); cd(1,1) -bd(1,1)];
21     b2 = w4^2 * cd - w2^2 * ab - w3^2 * bd;
22     q = inv(b1)*b2;
23     a4 = q(1,1);
24     a3 = q(2,1);
25
26
27     %output
28     vx = -w2*ab(2,1) - w3*be(2,1)
29     vy = w2*ab(1,1) + w3*be(1,1)
30     ax = -w2^2 * ab(1,1)-w3^2 * be(1,1) - a3*be(2,1)
31     ay = -w2^2 * ab(2,1)-w3^2 * be(2,1) + a3*be(1,1)
32

```

vx = 3.1487
 vy = 69.7278
 ax = 565.3788
 ay = -331.6450

Fig.22 Displacement, Velocity, and Acceleration Analytical values at $\theta = 180$ degree using MATLAB code

```

1  %input at 270 degree
2  a = [131.08;272.34];
3  c = [-66.92; 512.34];
4  b = [131.05;180.39];
5  d = [-62.92; 277.62];
6  e = [-170.75;283.64];
7
8  ab = b-a;
9  bd = d-b;
10 cd = d-c;
11 be = e-b;
12 w2 = 2;
13 %velocity
14 a1 = [cd -bd];
15 a2 = w2*ab;
16 z = inv(a1)*a2;
17 w4 = z(1,1);
18 w3 = z(2,1);
19 %acceleration
20 b1 = [-cd(2,1) bd(2,1); cd(1,1) -bd(1,1)];
21 b2 = w4^2 * cd - w2^2 * ab - w3^2 * bd;
22 q = inv(b1)*b2;
23 a4 = q(1,1);
24 a3 = q(2,1);
25
26
27 %output
28 vx = -w2*ab(2,1) - w3*be(2,1)
29 vy = w2*ab(1,1) + w3*be(1,1)
30 ax = -w2^2 * ab(1,1)-w3^2 * be(1,1) - a3*be(2,1)
31 ay = -w2^2 * ab(2,1)-w3^2 * be(2,1) + a3*be(1,1)
32

```

vx = 185.6148
 vy = 4.9523
 ax = -118.5036
 ay = 20.7910

Fig.23 Displacement, Velocity, and Acceleration Analytical values at $\theta = 270$ degree using MATLAB code

Kinematic parameter of output link (EH) at point E which are calculated above is summarized in table below .

Table 5: Summarized Analytical evaluation for Point E

Input Angle (degree)	x mm	y mm	vx mm/s	vy mm/s	ax mm/s ²	ay mm/s ²
0	95.47	254.43	15.6330	--94.1072	-120.64	-66.34
90	141.54	198.63	-184.507	1.0002	-427.408	335.577
180	279.85	276.53	3.1487	69.7278	565.38	-331.64
270	463.76	178	185.614	4.935	-118.50	20.7910

3.2: COMPARISON OF KINEMATIC PARAMETERS

3.2.1: The four values of displacement evaluated in section 3.1 superimposed in the plot of section 2.1.1

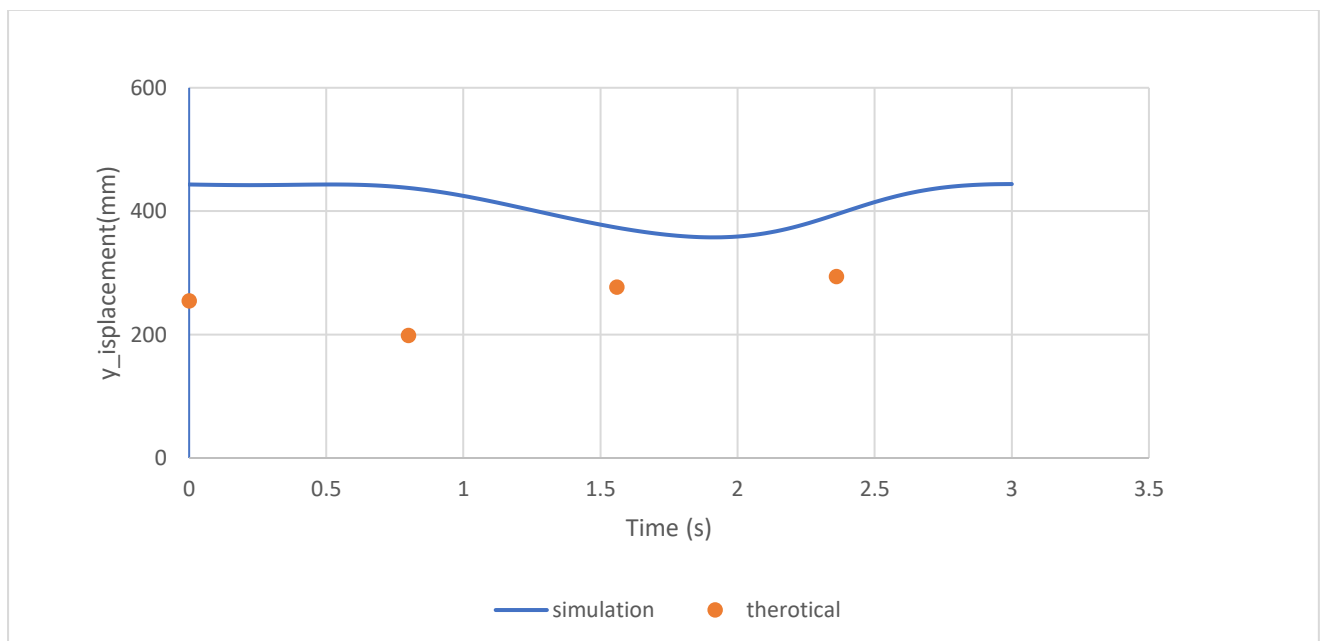
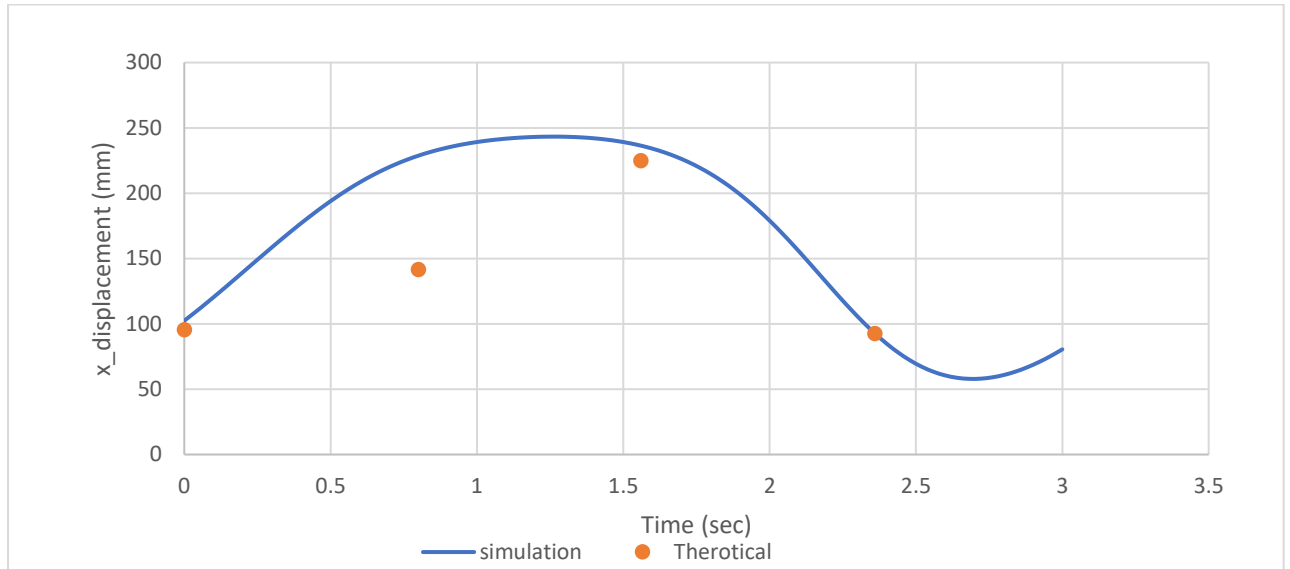


Fig.24 (Above) X-component of displacement and (Below) Y-component of displacement

3.2.2 :The four values of velocity evaluated in section 3.1 superimposed in the plot of section 2.1.2

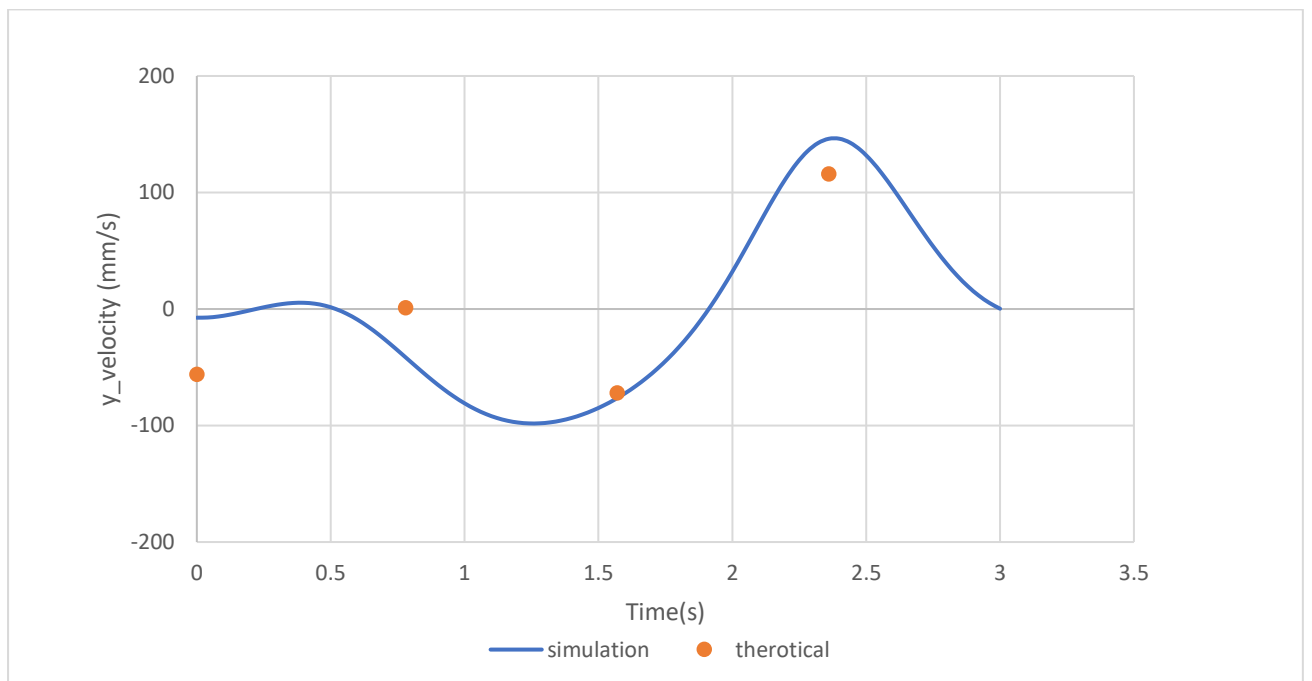
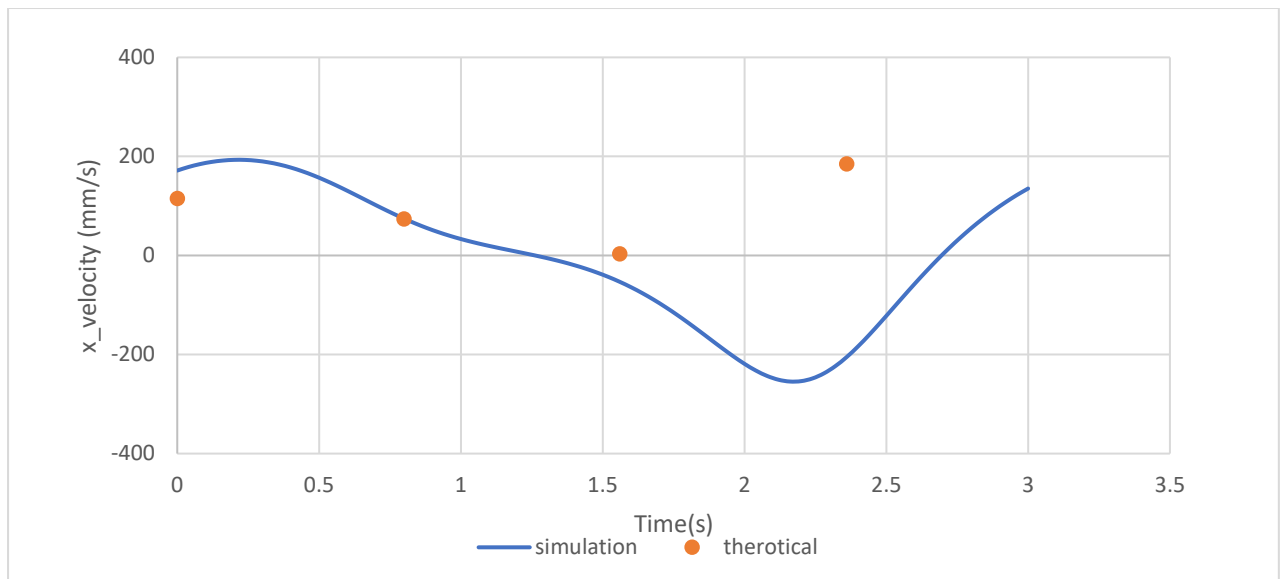


Fig.25 (Top) X - comp of Velocity and (Bottom) Y – comp of Velocity

3.2.3: The four values of acceleration evaluated in section 3.1 superimposed in the plot of section 2.1.3

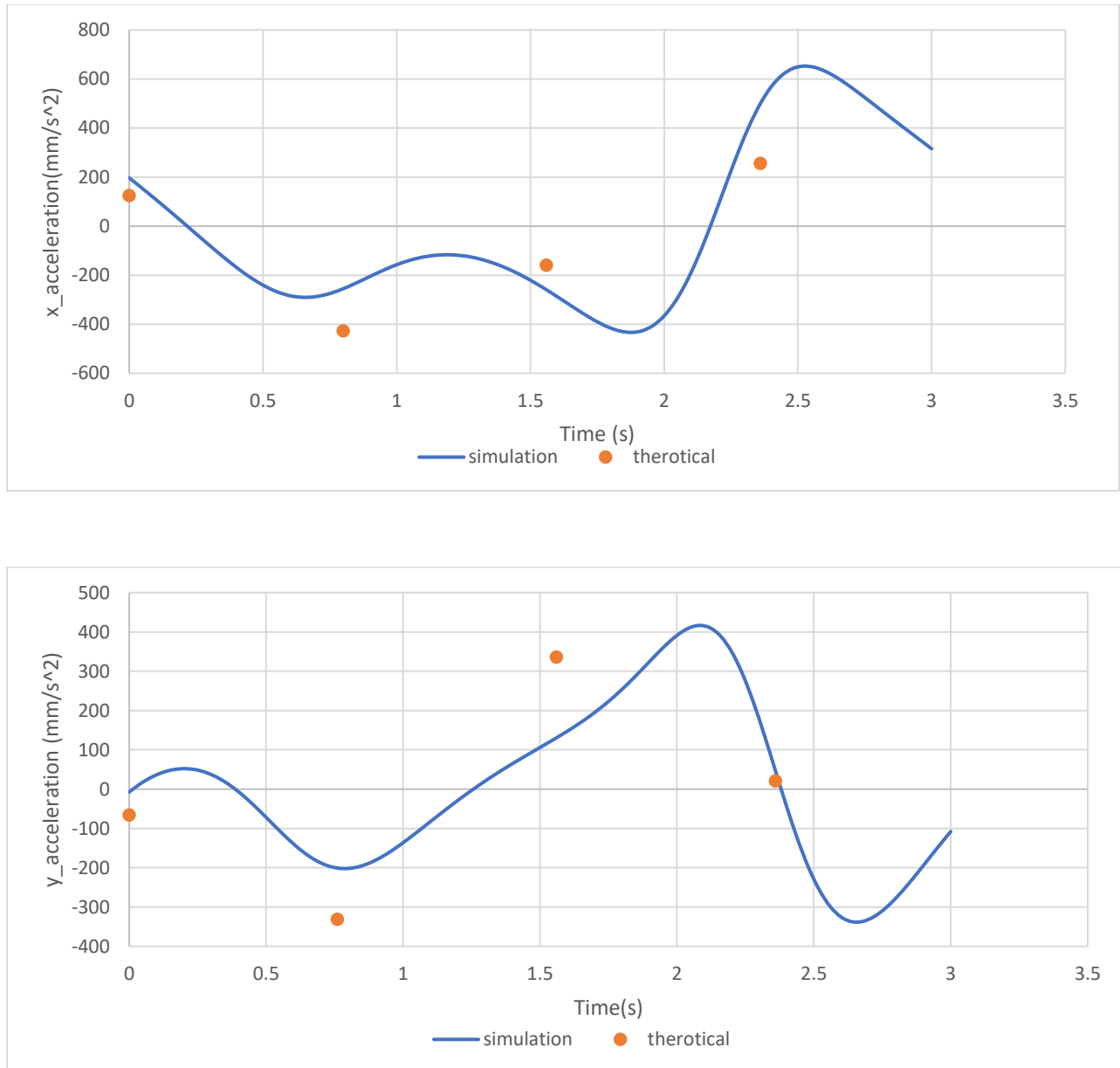


Fig. 26 (Top) X - comp of Acceleration and (Bottom) Y – comp of Acceleration

3.2.4 COMMENTS:

Reason for difference between Simulated and Analytical values:

- Since we are not taking account of angular velocity of output link EH and coupler link FG due to this there is little error in simulated and theoretical value.
- And other reason also there is difference between the crank position we have taken while calculating kinematic parameter theoretical is different than that of crank position of assembly of solid work.
- From graphical we find the position of output link at point E at different position of crank angle and we observed there is difference between simulated and the theoretical value and these value is further use to calculate velocity and acceleration of output link due to this error in velocity and acceleration diagram is magnified.

References

- [1] <https://www.youtube.com/watch?v=q6JKnxX6CtY>
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- [6] Overhead crane
<http://www.nucleoncranegroup.com/overhead-crane-accessories.html>
- [7] Forklift
<https://www.indiamart.com/proddetail/yellow-forklift-truck-22980908091.html>
- [8] Manual Trolley
<https://www.indiamart.com/proddetail/manual-trolley-1181723912.html>
- [9] Belt Conveyor
<https://sreevinayagaautomation.com/flat-belt-conveyor.php>

Box Transporting Mechanism

Self Project

by

Venkatesh Shukla
[25M0036]



Contents

0: BRIEF INTRODUCTION (FROM TERM PROJECT 1 SECTION 1.1)	3
1: STATIC FORCE ANALYSIS	3
2: VALIDATION OF STATIC FORCE ANALYSIS	6
3: DYNAMIC MOTION ANALYSIS.....	15
3.1 Constant driving force/torque.....	15
3.1.1: Plots for 20 N-mm constant torque	15
3.1.2 Superimposed plots for 40 N-mm with plots for 20N-mm	18
3.2 Sinusoidal driving force/torque.....	21
References	24

0: BRIEF INTRODUCTION (FROM TERM PROJECT 1 SECTION 1.1)

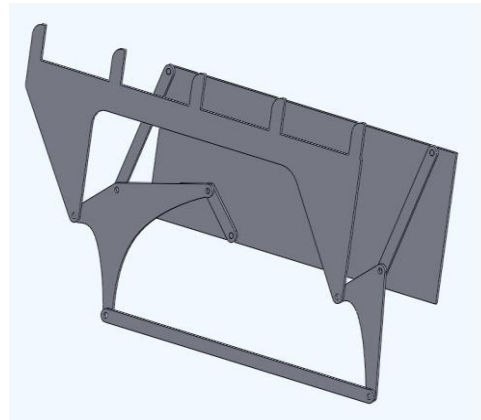


Fig.1 Actual Box Shifting Mechanism [1]

Box transport mechanism convert rotary motion of crank in reciprocating motion of carrier link it gives stop and move type of motion motion which is use in small scale industries for transporting boxes from one position to other.

1: STATIC FORCE ANALYSIS

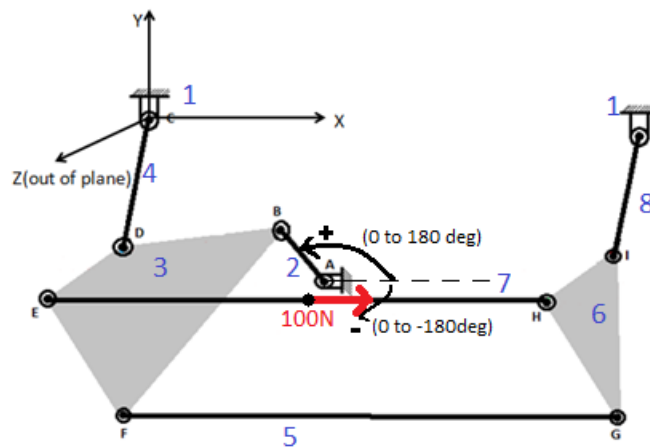


Fig.2 Mechanism Schematic with sign convention and link numbering

Magnitude of force= 100 N

Point of application =At the centre of mass of output link EH

And we do force analysis at four position of crank at (0 ,90, -90,180) degree to maintain equilibrium torque should be applied at input of crank to achieve static equilibrium of mechanism at four position of crank the simulated torque is obtained from solid work at different position of crank (0 .90 -90 180) degree .

The simulated static torque obtained corresponding to angle of **0 degree and 90 degrees, 180 degrees, -90 degrees**, which is to be applied on the input link are **-985.23 N-mm, 9723.55 N-mm, -184.23 N-mm, 9205.655 N-mm respectively** using Solid works software.

The negative value means that torque is applied around negative z global direction and positive torque value means it is applied around positive z global direction. And the direction of force is remains constant for all position of crank and snapshots corresponding to each 4 positions of crank shown below.

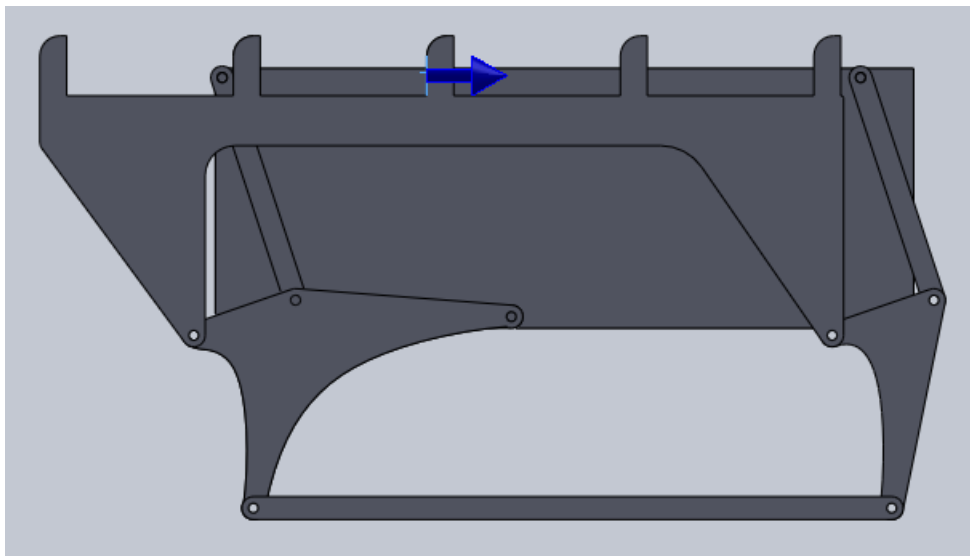


Fig.3 Crank Angle Position: 0 deg. Corresponding Simulated Torque: -985.23N-mm.

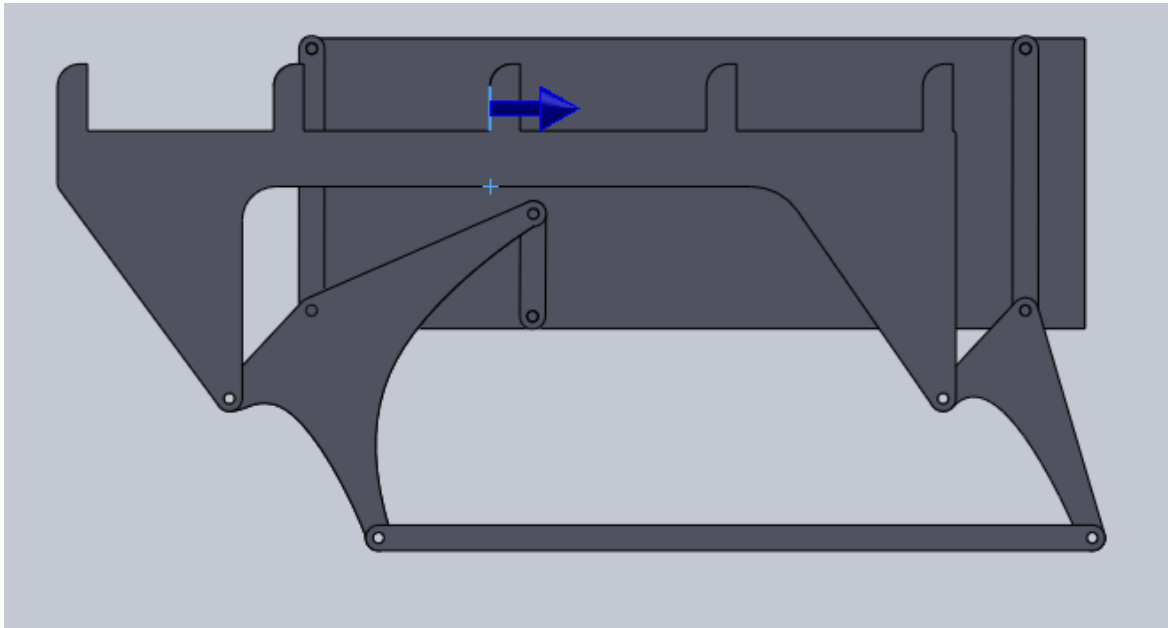


Fig.4 Crank Angle Position: +90 deg. Corresponding Simulated Torque: 9723.55 N-mm.

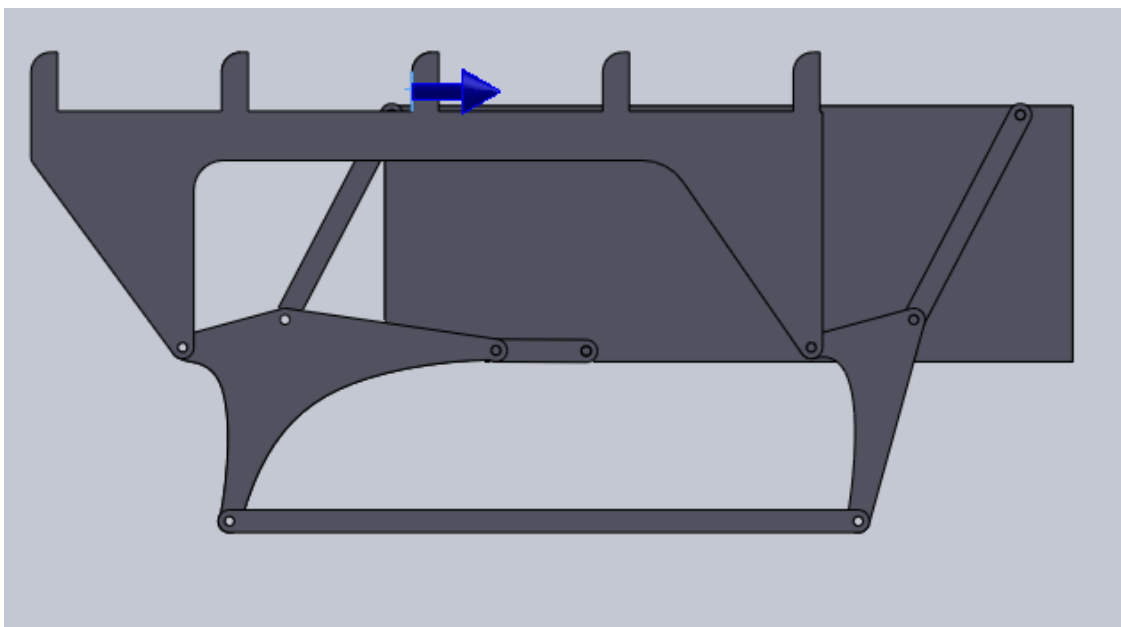


Fig.5 Crank Angle Position: 180 deg. Corresponding Simulated Torque: -184.23N-mm

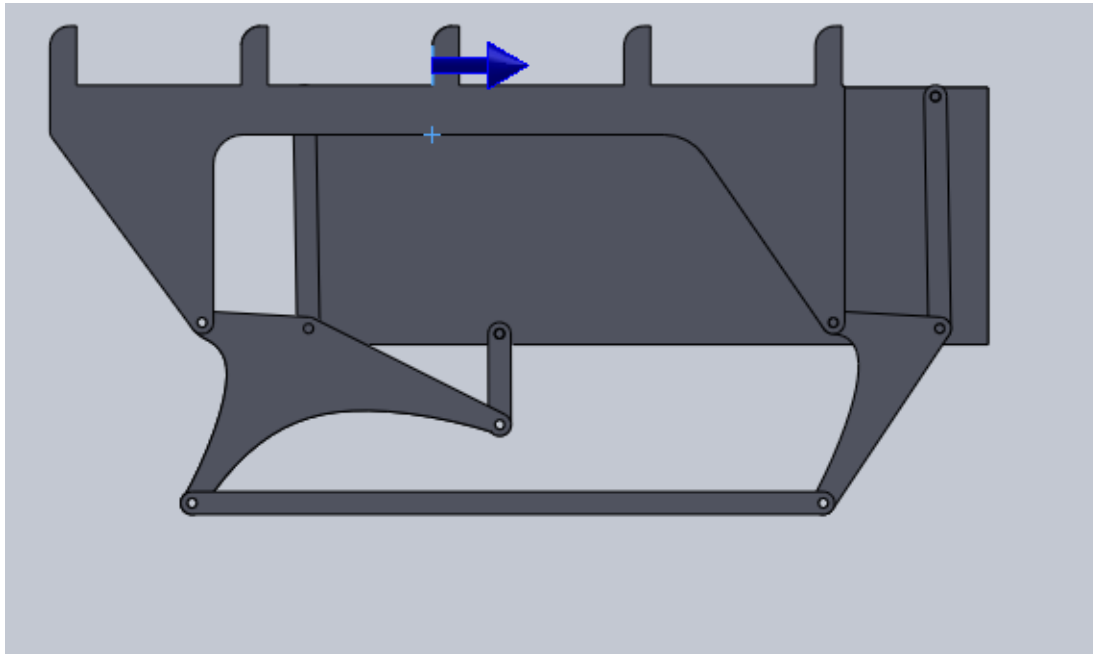


Fig.6 Crank Angle Position: - 90 deg. Corresponding Simulated Torque: 9205.655 N-mm.

2: VALIDATION OF STATIC FORCE ANALYSIS

Here main aim of this section is to find torque at the crank which is required to keep the mechanism in equilibrium when a constant force of 100 N is applied in positive x axis here we are not taking account of gravity to simplify the solution here mechanism is running in horizontal and vertical plane we are taking horizontal plane taken for calculation in vertical plane gravity acts while in horizontal plane weight of links acts downward in a direction perpendicular to horizontal plane which will not produce any moment

Notation used F_{pqr} =force applied by link p on q and r is the x and y component of force and i , and j is the unit vectors in x and y direction respectively and r_{bc} is the position vector of b on c

FBD of all links with reaction forces on joint shown below all position vector are taken from solid work model. here static force analysis is carried out for 90 degree a constant force of 100 N is applied at the centre of mass of output link.

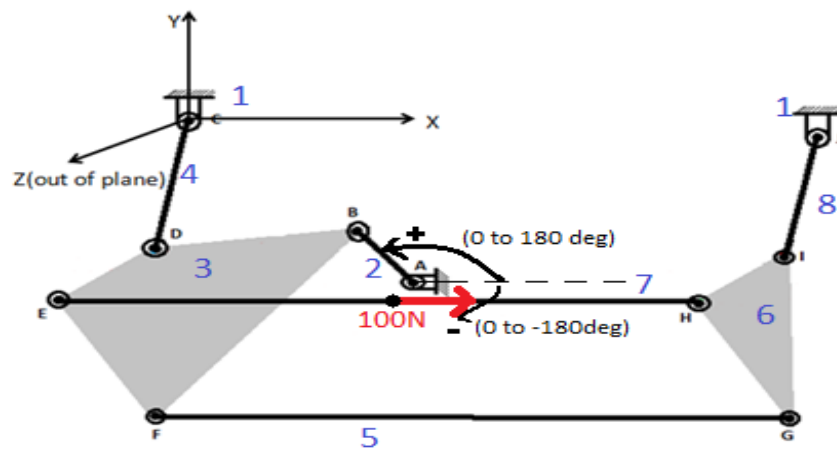
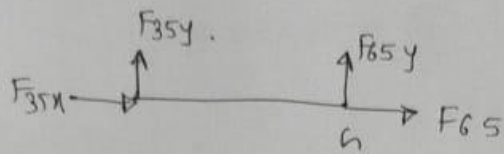
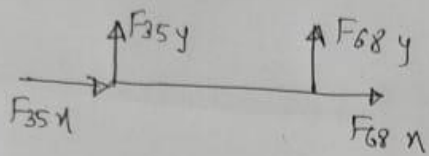


Fig .7: Mechanism Schematic with sign convention

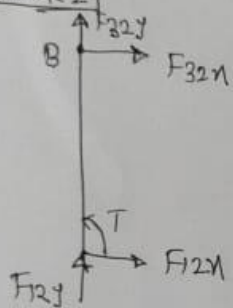
Link 5



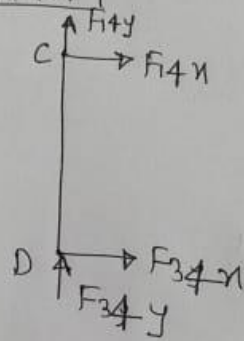
Link 8



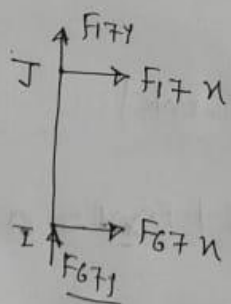
Link 2



Link 4

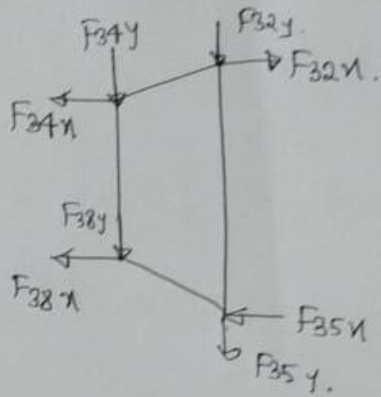


link 7

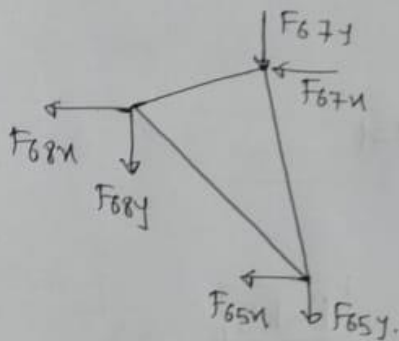


link 3

link 3



link 6



Link 8

for $\theta = -90^\circ$ (solid work)

$$\sum F_x = 0 ; \Rightarrow +100 + F_{38x} + F_{68x} = 0 \rightarrow (1)$$

$$\sum F_y = 0 ; \Rightarrow F_{38y} + F_{68y} = 0 \rightarrow (2)$$

$$\sum T_E = 0 ; (r_{EH1}) \times (-100\hat{i}) + (r_{EH2}) \times (F_{68x}\hat{i} + F_{68y}\hat{j}) = 0$$

$$r_{EH} F_{68y} = 0$$

$$F_{68y} = 0 \rightarrow (3) \text{ as } r_{EH} \neq 0$$

Fig. 8 Free body diagram of all the links

from eq (2)

$$F_{38y} = 0 \rightarrow (4)$$

Link 5

$$\sum F_x = 0; \Rightarrow F_{35x} + F_{65x} = 0 \rightarrow (5)$$

$$\sum F_y = 0; F_{35y} + F_{65y} = 0 \rightarrow (6)$$

$$\sum \tau_P = 0; (r_{P6} \hat{i}) \times (F_{65x} \hat{i} + F_{65y} \hat{j}) = 0$$

$$\boxed{F_{65y} = 0} \rightarrow (7)$$

from eq (6)

$$\boxed{F_{35y} = 0}$$

Link 6

$$\sum F_x = 0; -F_{67x} - F_{68x} - F_{65x} = 0 \rightarrow (9)$$

$$\sum F_y = 0 \Rightarrow -F_{67y} - F_{68y} - F_{65y} = 0$$

using eq (3) & (7)

$$\boxed{F_{67y} = 0}$$

$$\sum T_H = 0;$$

$$(\gamma_{H6x} \hat{i} - \gamma_{H6y} \hat{j}) \cdot (-F_{65x} \hat{i} - F_{65y} \hat{j}) \\ + (\gamma_{H7x} \hat{i} + \gamma_{H7y} \hat{j}) \cdot (-F_{67x} \hat{i} - F_{67y} \hat{j}) = 0$$

$$- \gamma_{H6x} F_{65y} \hat{k} + \gamma_{H6y} F_{65x} (-\hat{k}) + (-\gamma_{H7x} F_{67y} \hat{k} \\ + \gamma_{H7y} F_{67x} \hat{k}) \\ \Rightarrow -\gamma_{H6x} F_{65y} \hat{k} + \gamma_{H6y} F_{65x} (-\hat{k}) \\ + (-\gamma_{H7x} F_{67y} \hat{k} \\ + \gamma_{H7y} F_{67x} \hat{k}) = 0$$

Using eqⁿ (6) & (7)

$$\gamma_{H7y} F_{67x} - \gamma_{H6y} F_{65x} = 0 \rightarrow (11)$$

$$\boxed{11 < 7}$$

$$\sum F_x = 0; \Rightarrow F_{67x} + F_{17x} = 0 \rightarrow (12)$$

$$\sum F_y = 0; \Rightarrow F_{67y} + F_{17y} = 0 \rightarrow (13)$$

$$\sum T_E = 0; (\gamma_{E7} \hat{j}) \cdot (F_{17x} \hat{i} + F_{17y} \hat{j}) = 0$$

$$\gamma_{E7} F_{17x} = 0$$

$$\boxed{F_{17x} = 0}$$

from eqⁿ (12)

$$F_{67x} = 0 \rightarrow (15)$$

Using eqⁿ (15) & (16) we get from eqⁿ (9)

$$\boxed{F_{68} = 0} \rightarrow (17)$$

link 4

$$\sum F_x = 0 ; \quad F_{34x} + F_{14x} = 0$$

$$\sum F_y = 0 ; \quad F_{34y} + F_{14y} = 0 \rightarrow (18)$$

$$\sum T_D = 0 ; \quad (r_{DCj}) \times (F_{14x} \hat{i} + F_{14y} \hat{j}) = 0$$

$$r_{DC} F_{14x} = 0$$

$$\boxed{F_{14x} = 0} \rightarrow (19)$$

Using eqⁿ (19)

$$F_{34x} = 0 \rightarrow (20)$$

link 3

$$\sum F_x = 0 ; \quad F_{32x} + F_{34x} + F_{38x} + F_{35x} = 0 \rightarrow (21)$$

$$\sum F_y = 0 ; \quad F_{32y} + F_{34y} + F_{38y} + F_{35y} = 0 \rightarrow (22)$$

$$\sum \vec{r} \times \vec{F} = 0;$$

$$(\gamma_{ED} x \hat{i} + \gamma_{ED} y \hat{j}) \times (-F_{34} x \hat{i} - F_{34} y \hat{j}) +$$

$$(\gamma_{EB} x \hat{i} + \gamma_{EB} y \hat{j}) \times (-F_{32} x \hat{i} - F_{32} y \hat{j}) +$$

$$(\gamma_{EF} x \hat{i} - \gamma_{EF} y \hat{j}) \times (-F_{35} x \hat{i} - F_{35} y \hat{j})$$

$$= -\gamma_{ED} x F_{34} y - \gamma_{EB} x F_{32} y + \gamma_{EB} y F_{32} x = 0$$

→ (23)

Using eqⁿ (17) & eqⁿ (1) becomes

$$\boxed{F_{38} x = 100} \rightarrow (24)$$

Using eqⁿ (20) & $\boxed{F_{35} x = 0}$ & from eqⁿ (21)

$$\boxed{F_{32} x = +100 \text{ N}}$$

$$r_{EDn} = 104.23 \text{ mm} ; r_{EBn} = 312.82 \text{ mm} \quad r_{EB} = 218.42$$

$$(104.23)F_{32y} - (312.82)F_{32y} + (218.42)F_{32x} = 0$$

$$\boxed{F_{32y} \Rightarrow 104.712 \text{ (N)}}$$

link ②

$$T_K + r_{ABj} \times (F_{32n}\hat{i} + F_{32j}\hat{j}) = 0$$

$$T_K - 91.95 \times 100 = 0 \Rightarrow ; \boxed{T_K = 9195 \text{ (Nmm)}}$$

Error Analysis: Torque obtain by solid work simulation is 9205.655N mm and from our analytical calculation its is 9195 N mm error is around 0.11% which may be arise due to manual calculation or measuring position vectors.

3: DYNAMIC MOTION ANALYSIS

3.1 Constant driving force/torque

Here we give torque of 20 N-mm on crank AB and there is increase of 40 % velocity magnitude from first cycle to second cycle. For all possible values of constant torque, it can be observed that the amplitude of velocity increases by 30-60%. If we want the velocity amplitude increases by 15-20 % we need to change the dimension of link. Velocity and Acceleration of point E (same as in term project1) is plotted for both constant torque (20 N-mm) in section 3.1.1 and double constant torque. (40 N-mm) in section 3.1.2.

3.1.1: Plots for 20 N-mm constant torque

Crank AB is subjected to constant torque of 20 N mm in counter clockwise for 3.8 sec to complete two revolution and here as we can see velocity and acceleration is constant for some time and we can see from fig 9 the first cycle take more time compared to second cycle because of constant torque input and we can see velocity and acceleration magnitude is increases from second cycle onwards for both x and y component .

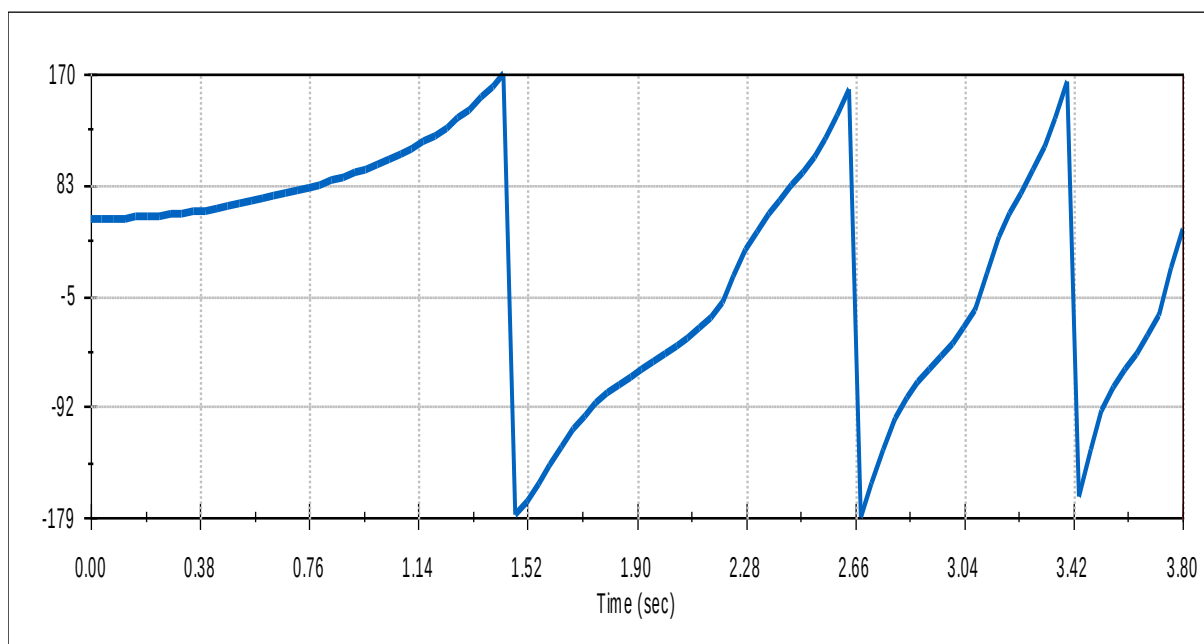


Fig.9 Angle of input link AB with respect to time for its two complete revolutions

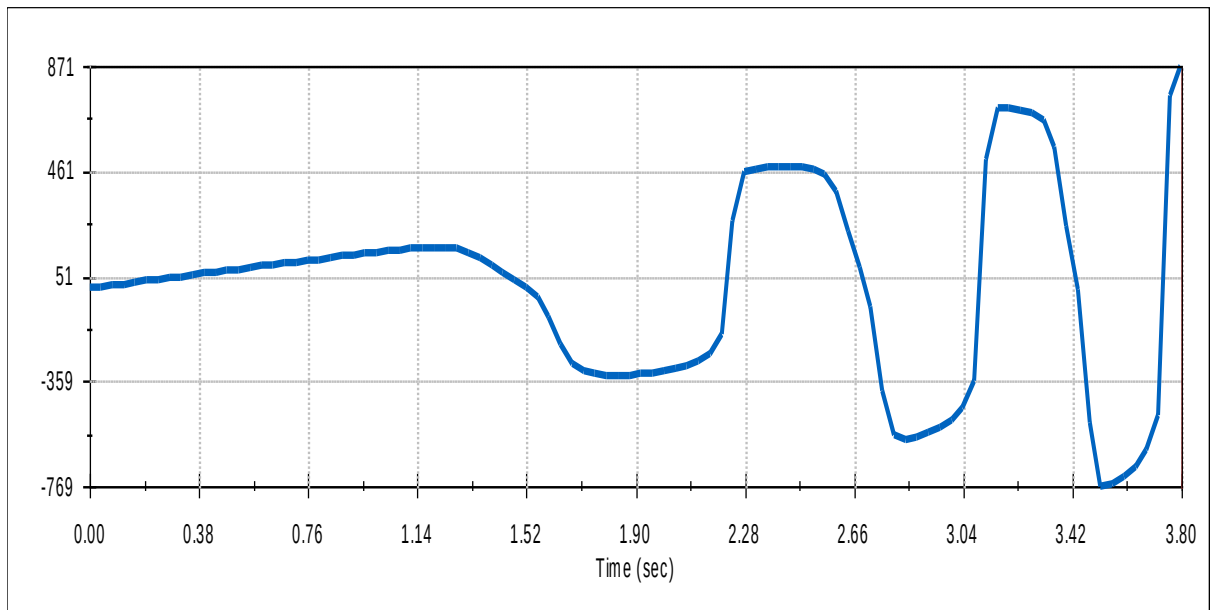


Fig.10 Velocity x-component for constant torque (20 N-mm)

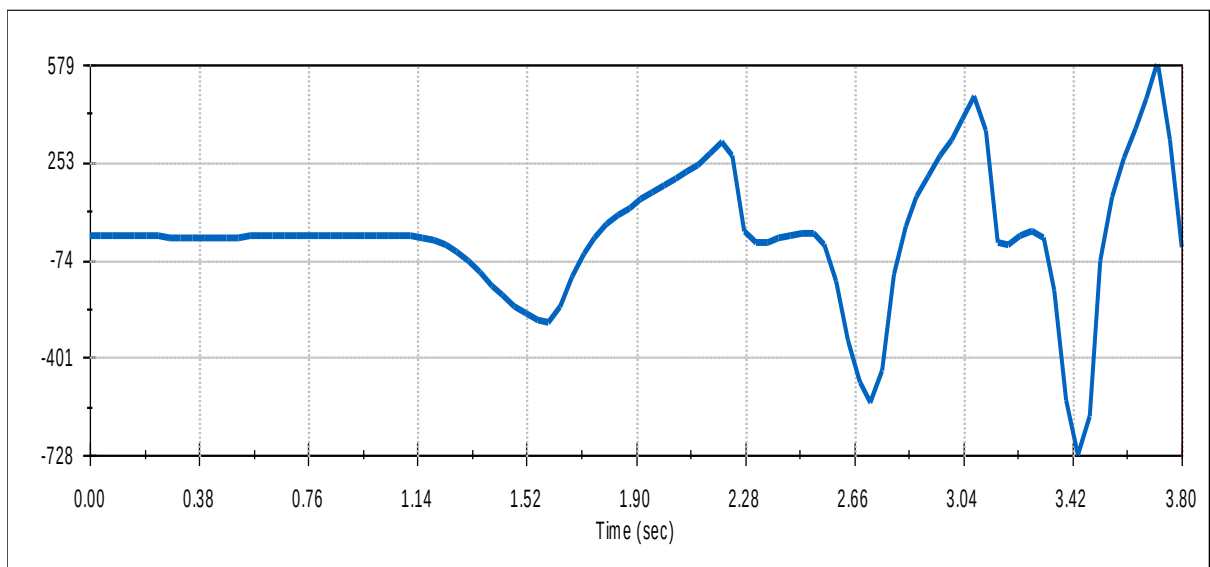


Fig.11 Velocity y-component for constant torque (20 N-mm)

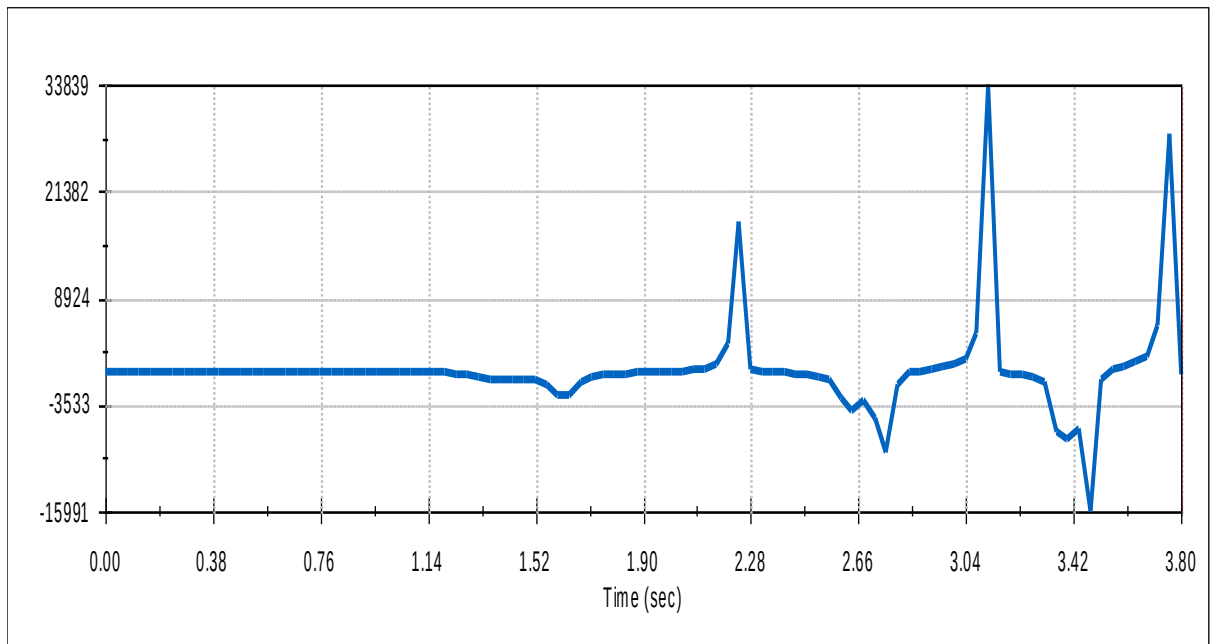


Fig.12 Acceleration x-component for constant torque (20 N-mm)

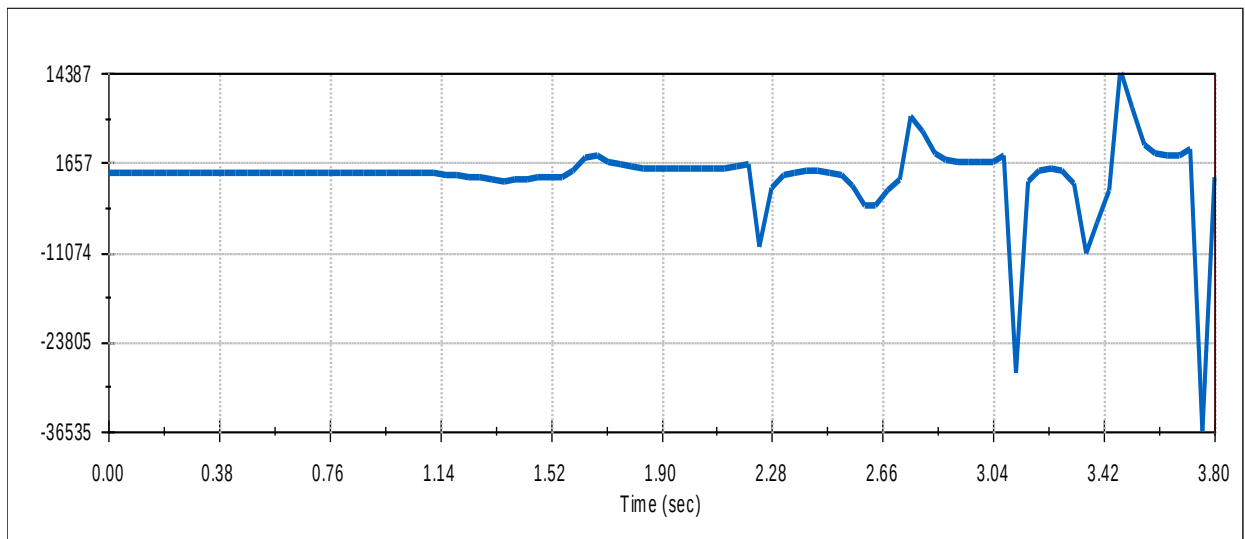


Fig.13 Acceleration y-component for constant torque (20 N-mm)

3.1.2 Superimposed plots for 40 N-mm with plots for 20N-mm

In this section crank (driver) which is subjected to 40 N-mm which is double that of what is imposed on section 3.1.1 when crank is subjected to 40 N mm it complete its two cycle at 3.16 sec superimposed plot for 2 cycle is shown below that is for 20 N-mm and 40 N -mm torque .

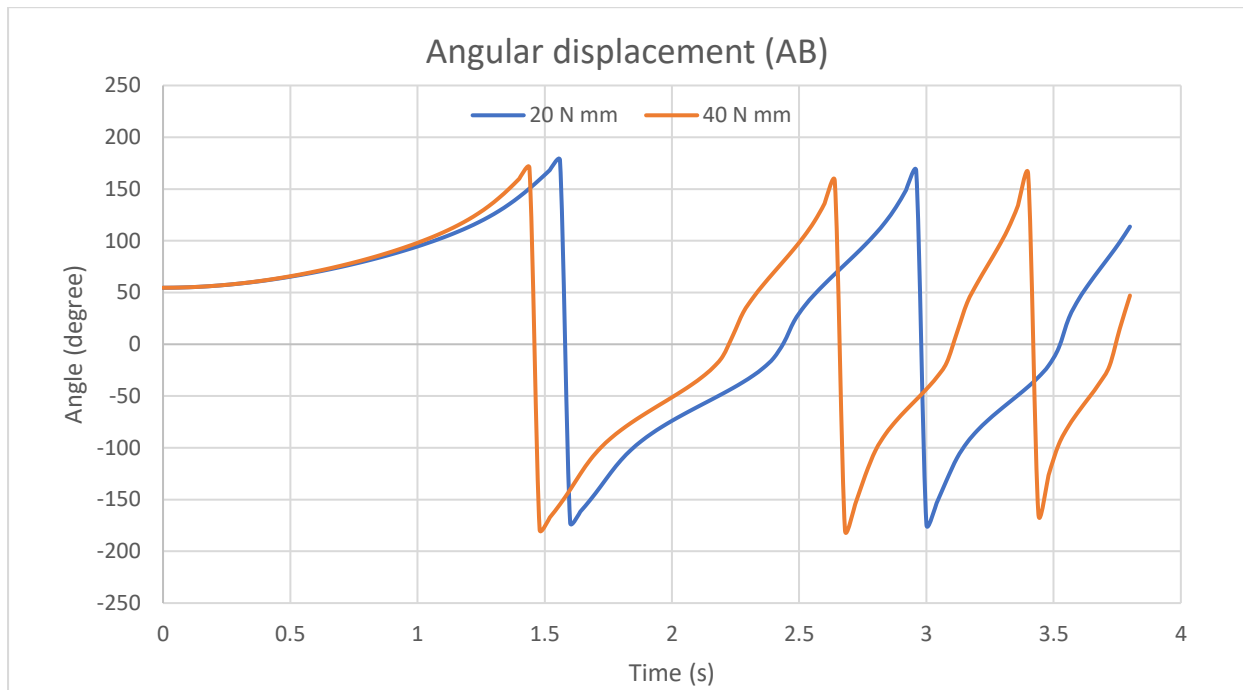


Fig.14 Angle of input link AB with respect to time for its two complete revolutions

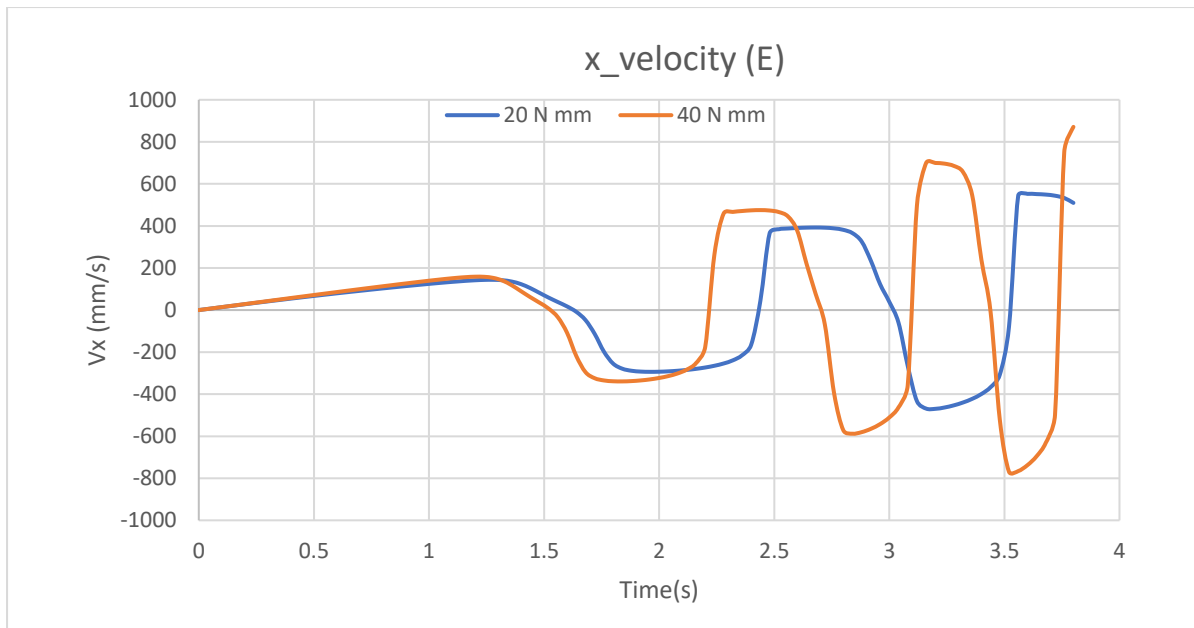


Fig.15 Velocity x-component of point E with respect to time for its two complete revolutions.

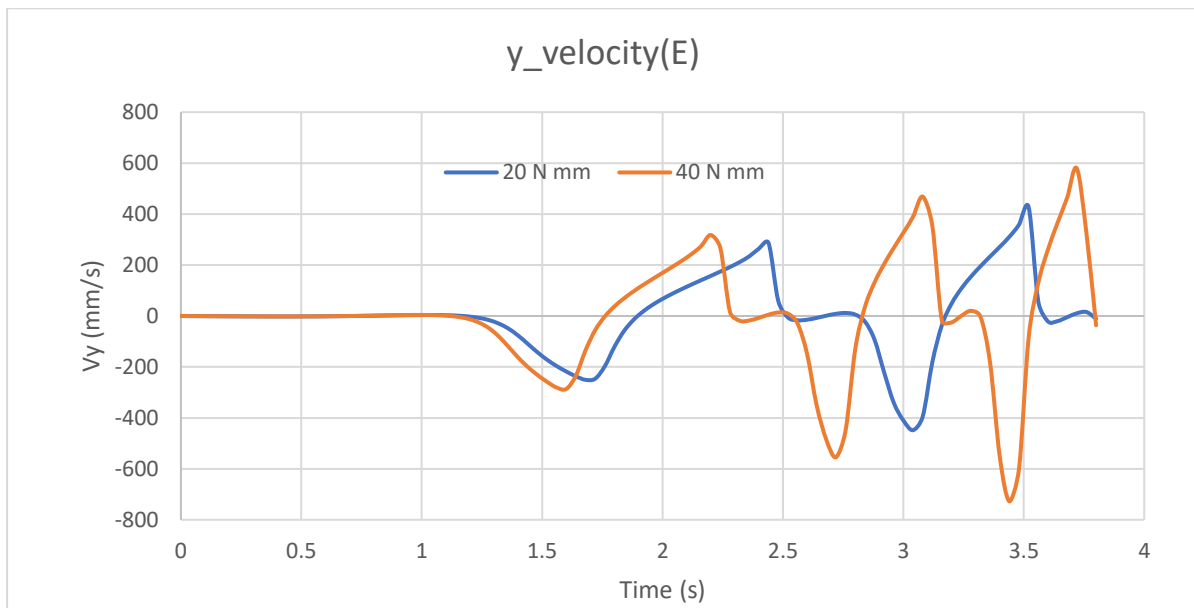


Fig.16 Velocity y-component of point E with respect to time for its two complete revolutions

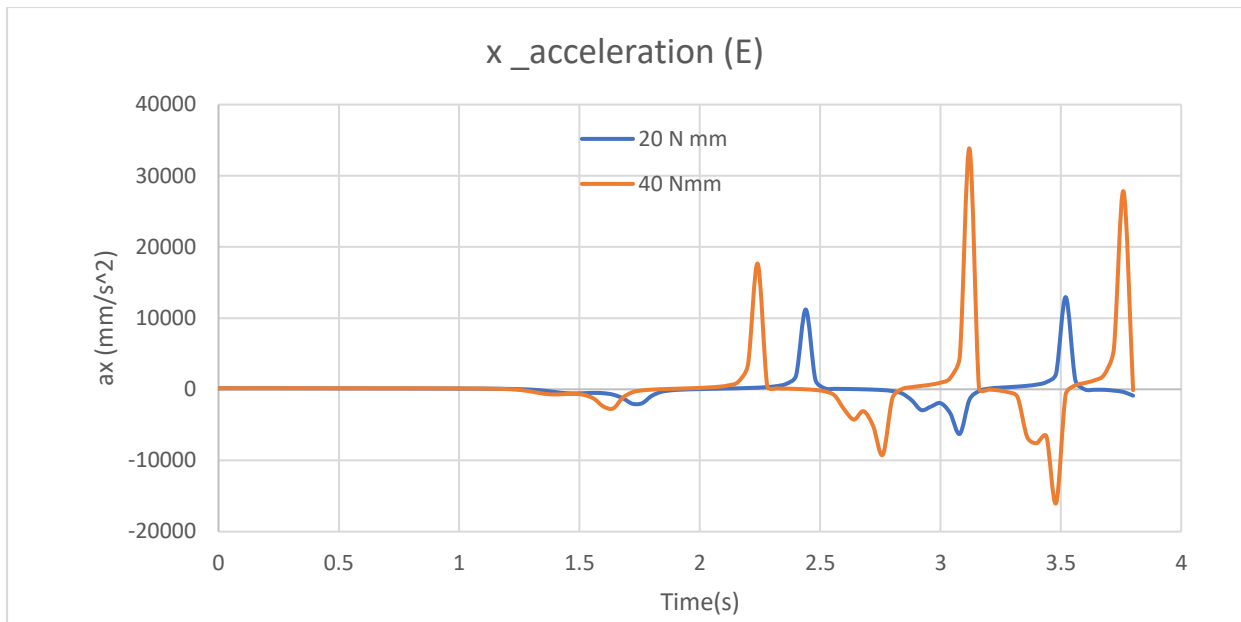


Fig.17 Acceleration x-component of point E with respect to time for its two complete revolutions.

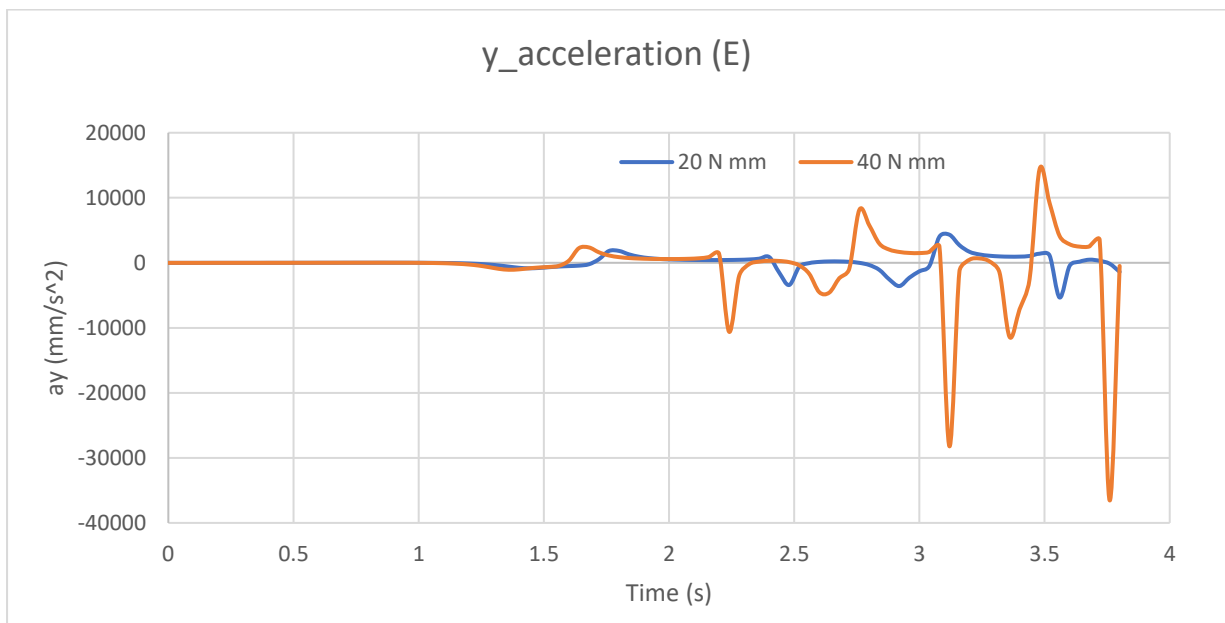


Fig.18 Acceleration y-component of point E with respect to time for its two complete revolutions

3.2 Sinusoidal driving force/torque

In this section we applied sinusoidal torque to crank i.e. $(28+28*\sin(t))$ N-mm and amplitude of sinusoidal torque is 1.4-time 20 N-mm and as we can see in fig 19 the minimum value of torque is zero. velocity and acceleration graph are plotted against time for point E of output link carrier (EH) for two cycles.

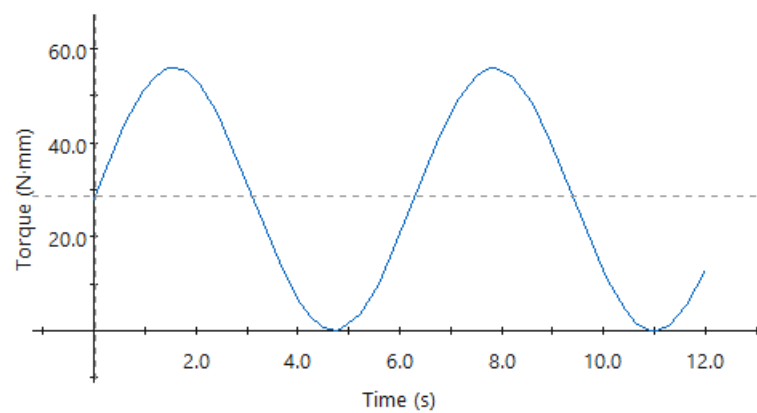


Fig.19 Torque ($T = 28 + 28 \sin(t)$) Vs time

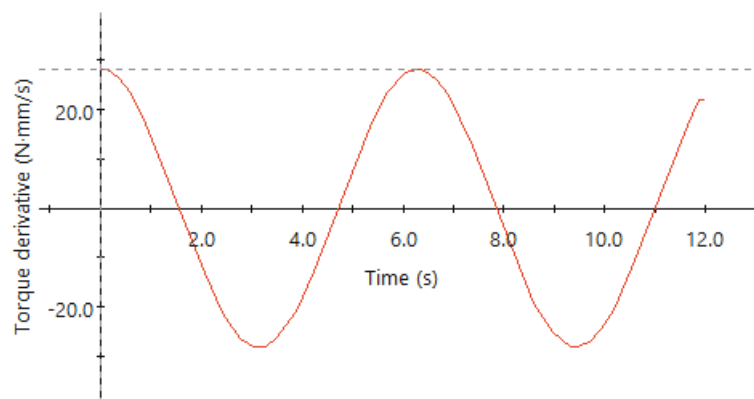


Fig.20 Torque derivative Vs time

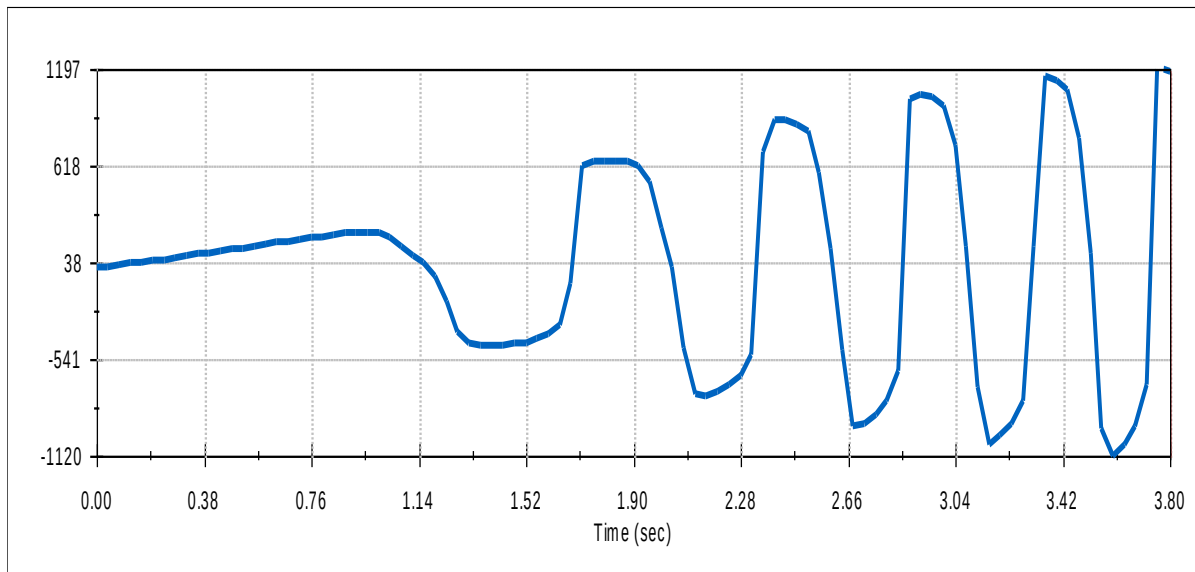


Fig.21 Velocity x-component of point E with respect to time for its two complete revolutions

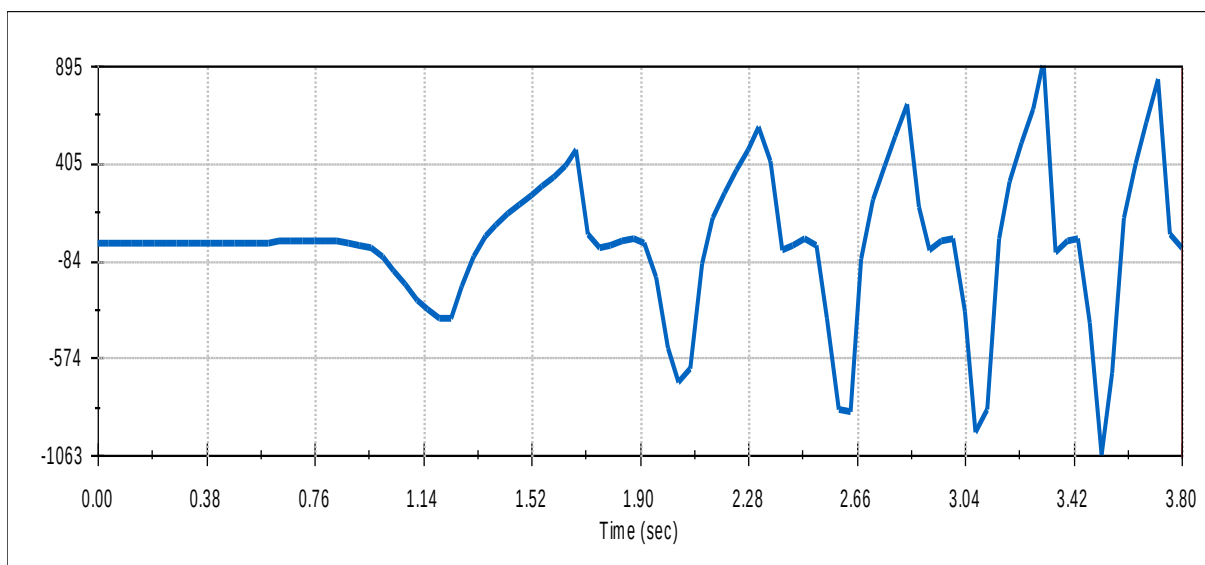


Fig.22 Velocity y-component of point E with respect to time for its two complete revolutions.

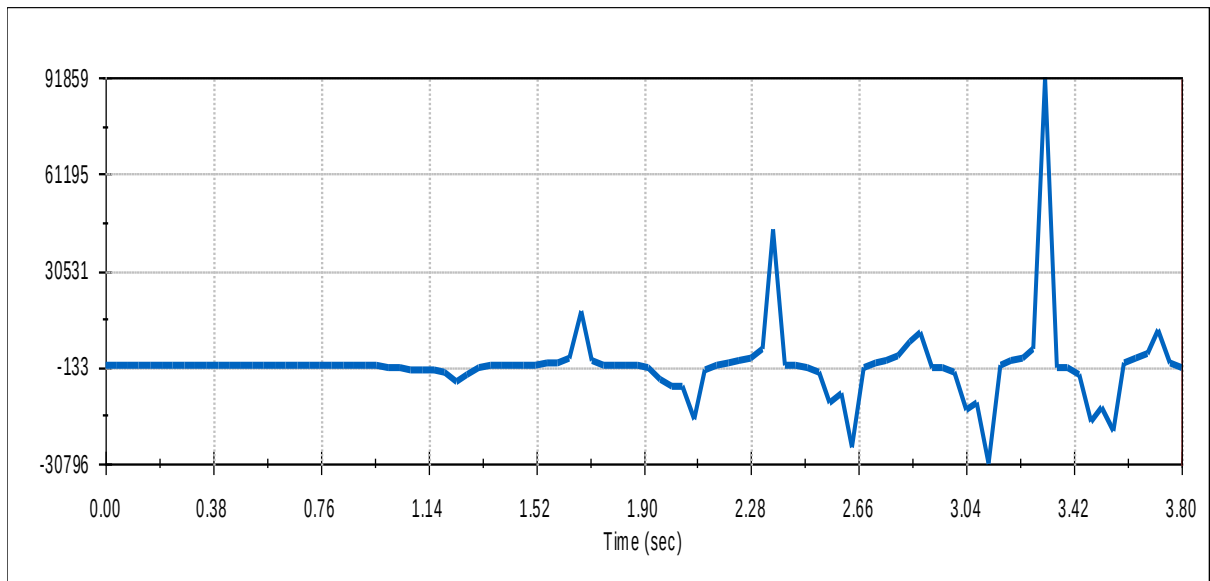


Fig.23 Acceleration x-component of point P with respect to time for its two complete revolutions.

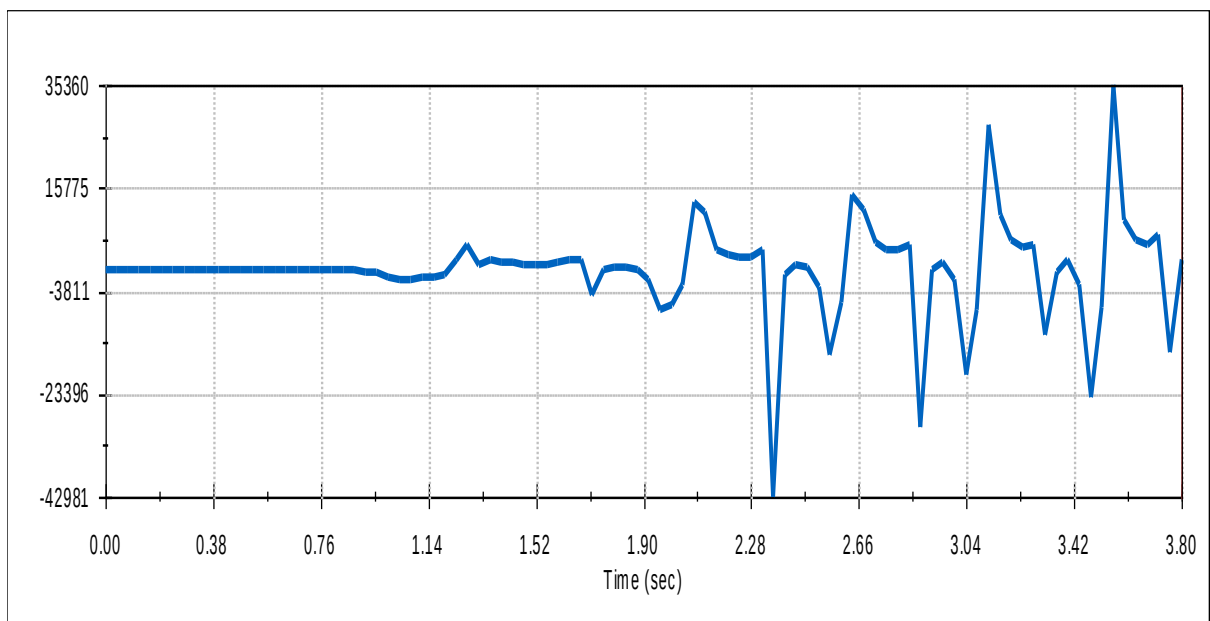


Fig.24 Acceleration y-component of point P with respect to time for its two complete revolutions.

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